

Gold Trend Champion — Complete Technical Archive

Machine-readable specification and performance record of every formal version of the Gold Trend strategy. Every parameter is stated, not described. Where a figure does not exist it reads NOT MEASURED. Rejected experiments are excluded by instruction.

Generated 2019-12-01 data through 2026-07-30. Build date 2026-08-05.

Current champion

Current champion is Config G1*. It is preferred over G1 because at a matched 25% drawdown budget it more than halves the Monte Carlo tail — P(DD>30%) 19.28% to 6.48% — while raising profit factor 2.005 to 2.157 and pre-2025 profit factor 1.513 to 1.600, at equal MAR (2.510 vs 2.511). G1 was preferred over E for the same reason at smaller magnitude. E was preferred over the Original Champion on every metric simultaneously.

Versions in this document

#	Version	Pine source	Shipped?
1	Original Champion	git 5388d74:strategies/gold_trend_strategy.pine	Yes (historic)
2	Config E	git 927f485:strategies/gold_trend_E.pine	Yes (historic)
3	Config G1	git 927f485:strategies/gold_trend_G1.pine	Yes (historic)
4	Config G1*	strategies/gold_trend_G1.pine (HEAD, 7b25357)	YES — CURRENT

Excluded, with reason: **gold_trend_core.pine** (commit 346e300) is a superseded development build, never a formal baseline. **gold_trend_trailing.pine** (commit 23ab656) is the exit-experiment harness, not a strategy version. All rejected experiments — volatility targeting, level filters, PWH break, ORB, Quarterly Theory overlays, Bollinger squeeze gate — are excluded by instruction.

Global — Data specification

Field	Value
Instrument	XAUUSD (spot gold), OANDA feed
Timeframe	30 minutes, resampled from a native 15-minute export
Source file	data/xauusd_15m.csv.gz (157,366 15m bars)
Bars (30m)	78,695
Date range	2019-12-01 23:00:00 to 2026-07-30 00:00:00 (UTC, tz-naive)
Span	6.66 years, 349 complete weeks
Price range	low 1,451.43 to high 5,598.09
Data quality	0 duplicate timestamps; 0 bars with high < low; 5 zero-range bars; 394 gaps > 2h (weekends). No price spikes: the
Regime coverage	Gold ran 1,451 -> 5,598 (Jan 2026) -> 4,081. The sample contains a genuine 27% top-to-current decline as well as
Signal evaluation	CLOSED BARS ONLY for every version.

Global — Research protocol

Field	Value
Drawdown budget solving	Risk% is an OUTPUT, not an input. Bisection over risk% in [0.02, 4.0], 18 iterations, accepting the largest risk whos
Monte Carlo	2,000-5,000 paths per seed x 5 seeds (11, 101, 2027, 55555, 987654). EQUITY-RELATIVE returns are shuffled, no
"Pre-2025"	All trades whose ENTRY bar falls in calendar year 2024 or earlier. This deliberately excludes the 2025-2026 trend,
TRAIN / VALID / TEST	The trade sequence split into equal thirds by trade index (not by time).
Anchored walk-forward	Six equal forward windows spanning bars 31,478 to 78,695 (the final 60% of the sample), expanding anchor.
Concentration	Trades sorted by P&L descending; the top 5 and top 10 expressed as a percentage of NET profit and of GROSS pr
Costs	Commission USD 0.07/contract/side and slippage 0.20 points/side applied in every figure in this document.

Global — Cross-version comparison, 25% drawdown budget

Metric	Original	E	G1	G1* (current)
Risk per trade (solved)	0.674%	0.706%	0.724%	0.646%
Trades	789	711	711	711
Long / short	504 / 285	467 / 244	467 / 244	467 / 244
Win rate	21.55%	21.66%	21.52%	21.66%
Average R	0.608	1.256	1.453	1.663
Profit factor	1.585	1.945	2.005	2.157
Net return	658.3%	1,878.5%	2,460.5%	2,463.7%
CAGR	35.56%	56.56%	62.75%	62.78%
Max drawdown	25.00%	25.00%	25.00%	25.00%
MAR	1.423	2.263	2.510	2.511
Monte Carlo median DD	26.29%	25.36%	24.86%	21.12%
MC 95th pct DD	38.03%	36.67%	35.95%	30.93%
P(DD > 30%)	27.42%	21.67%	19.28%	6.48%
PF pre-2025	1.258	1.459	1.513	1.600
PF TRAIN/VALID/TEST	1.28/1.20/1.89	1.49/1.27/2.27	1.55/1.35/2.29	1.71/1.49/2.47
Avg hold (bars)	43.6	40.7	40.8	40.7
Avg adds per trade	1.831	1.809	1.809	1.809
Top 5 trades, % of NET	42.1%	48.9%	47.6%	42.3%
Top 10 trades, % of NET	73.2%	81.4%	80.2%	70.7%

Global — Cross-version comparison, 20% drawdown budget

Metric	Original	E	G1	G1* (current)
Risk per trade (solved)	0.534%	0.553%	0.567%	0.506%
Profit factor	1.566	1.906	1.967	2.116
CAGR	28.02%	43.33%	47.84%	47.77%
MAR	1.401	2.166	2.392	2.389
Monte Carlo median DD	21.31%	20.36%	19.96%	16.87%
P(DD > 30%)	7.20%	4.88%	4.12%	0.85%
PF pre-2025	1.270	1.472	1.528	1.626
Top 5 trades, % of NET	38.9%	43.4%	41.5%	36.5%

Global — Known limitations and caveats

1. Profit concentration. At the 25% budget the top 5 of 711 trades (0.7%) carry 42.3% of NET profit for G1*, and the top 10 carry 70.7%. The strategy is a small number of large winners plus a long tail of small losses. This is inherent to trend following with wide trails and pyramiding, and it means live results depend heavily on catching those few moves.

2. Gold-only scope. US30 was withdrawn from the protocol by user directive on 2026-08-05. Before withdrawal, G1* FAILED the US30 transfer test: P(DD>30%) rose from 36% to 53%. The claim the evidence supports is "better on gold", not "better".

3. Execution sensitivity. Break-even is roughly 1.6 points of slippage. At 0.50 points returns fall sharply; at 1.50 points the system is dead. Overnight financing is not modelled at all (estimated 2-8% of equity per year).

4. Walk-forward fold 1 degrades for G1*. Fold 1 of 6 falls to PF 0.925; folds 2-6 improve on G1.

5. Whole-contract granularity. Below roughly USD 25,000 at current gold prices the integer-lot floor causes a large fraction of signals to be skipped and the system loses money. All research figures use FRACTIONAL sizing (frac_qty=True); the Pine uses math.floor and will therefore differ on small accounts.

6. Parameter selection is not virgin out-of-sample. Parameters were chosen across earlier rounds with knowledge of the full sample.

7. Never seen a multi-year gold bear market. The sample contains three down/flat years (2021, 2022, 2026) but no sustained bear market.

Global — Bugs discovered during research that affect interpretation

ID	Defect	Effect on earlier numbers
BUG-029	In backtest/orb.py, a bar that hit both target and stop fell through ORB session PF fell 1.18 ex 0.99 due to the fix. Caught by a strategy AND its	
BUG-030	Pyramid adds filled at a stale trigger price when the bar gapped Champion PF fell 1.646 -> 1.623 after the gap-aware fix. Caught by a +5.4 trillion p	
np.floor sizing	Integer position sizing silently rejected 788 of 789 US30 entries, Anytime in US30 figure is void. frac_qty flag added.	
stop_atr not passed	champion.py did not pass stop_atr to exit_lab, so the exit used a default while default figures 4.24e computed with a mismatched initial stop.	
risk_series not applied to units	did not rescaled the opening unit only. Correct for G1 by design. Discovered a volatility targeting (VST) measure reproducing exactly). This is w	
G1 constant mismatch	research/voltarget.py g1_series() used 1.30/0.85 at a 1.10 volatility. The result is the shipped previously published results. RSEUCFCAEDN and other fea	

Note on the G1 constant mismatch

The discrepancy favours the shipped code. Recomputed at a 25% drawdown budget, G1* with the PINE constants gives PF 2.157, MAR 2.511, P(DD>30%) 6.48%, pre-2025 PF 1.600. The previously published harness figures were PF 2.175, MAR 2.431, P(DD>30%) 9.85%, pre-2025 PF 1.520. The Pine is better on MAR, on the Monte Carlo tail and on the pre-2025 regime split. The earlier numbers were therefore conservative, but they did not describe the code that ships.

Version 1 — Original Champion

1. Identity and provenance

Field	Value
Label in research	ORIGINAL / "champion" (backtest/champion.py defaults)
Pine file	strategies/gold_trend_strategy.pine
Commit documented here	5388d74 — last commit at which the Pine matches backtest/champion.py's ORIGINAL constants
Validation certificate commit	d04158c (DSR 0.9996, PBO 0.099)
Change vs previous	N/A — this is the baseline.
Was it shipped/live?	Yes. This carried the early validation certificate.

2. Complete rule specification

2.1 Entry

Field	Value
Bar close only	Yes. Every condition is evaluated on the CLOSED bar. No intrabar evaluation, no varip, no request.security with loc
Donchian breakout lookback	24 bars = 12.0 hours on 30m. Pine: entryLenE = max(5, int(12.0*60/tfMin)). Applied with a [1] offset: donHi = ta.hig
Regime EMA	EMA(close, 1008 bars) = 21.0 days on 30m. Pine: regimeLenE = max(100, int(21.0*24*60/tfMin)). Bar-based, not c
Fast SMA	SMA(close, 384 bars) = 8.0 days on 30m. Pine: smaLenE = max(50, int(8.0*24*60/tfMin)).
Slow SMA	SMA(close, 1008 bars) = 21.0 days
Slope lookback	96 bars = 2.0 days on 30m. Pine: slopeLenE = max(20, int(2.0*24*60/tfMin)).
LONG conditions (all required)	1) close > ta.highest(high,24)[1] 2) close > EMA(close,1008) 3) close > SMA(close,384) 4) close > SMA(close,S
SHORT conditions (all required)	1) close < ta.lowest(low,24)[1] 2) close < EMA(close,1008) 3) close < SMA(close,384) 4) close < SMA(close,SL
Other entry filters	NONE. No session filter, no volume filter, no volatility filter, no candlestick or level confluence.
Cooldown after an exit	3 bars — a new entry requires barsSinceExit > cooldown.

2.2 Exit

Field	Value
Initial trail multiple	4.24 x ATR(14). Pine: trailMultE = 6.0 * sqrt(15.0/tfMin) = 4.2426 at tfMin=30.
Number of exits	Exactly one: a ratcheting chandelier trailing stop.
ATR period	ATR(14), Wilder smoothing (RMA / ewm alpha=1/14).
Tightening rule	NONE. The trail multiple is a constant 4.24 x ATR(14) for the whole trade.
Trail anchor	PREVIOUS bar's extreme. Long: high[1] - atr*mult. Short: low[1] + atr*mult. A bar can never move its own stop and
Ratchet	Monotone in the trade's favour only. Long: trailStop := max(trailStop, cand). Short: min(trailStop, cand).
Profit target	NONE.
Time exit	NONE.
Breakeven stop	NONE.
Max holding period	Unbounded.

2.3 Pyramiding

Field	Value
Maximum adds	4
Add trigger spacing	every 1.5 x ATR of favourable movement from the last add price
Add size formula	aQty = equity * (riskE * shortRiskIfShort * pyrRisk) / 100 / (ATR_CURRENT * 4.24). NOTE: the ORIGINAL sizes ad
Regime multiplier on adds	N/A — no regime multiplier exists in this version.
Open risk treatment	Unbounded by design. Measured median open risk ~2.1R, maximum >5R. Hard caps were tested and destroyed m

2.4 Position sizing and risk

Field	Value
Risk base	strategy.equity (mark-to-market equity, not balance)
Size formula	qty = floor(min(equity * effRisk / 100 / stopDist, equity * 20 / close))

Field	Value
stopDist	ATR(14) * 4.24 — the same distance as the initial trail
Short risk multiplier	0.75 (shorts risk 75% of the long risk)
Regime multiplier	NONE
Risk solved to 25% DD budget	0.674%
Risk solved to 20% DD budget	0.534%

2.5 Costs and execution assumptions used in research

Field	Value
Commission	USD 0.07 per contract (ounce) per side
Slippage	0.20 price points per side, applied against the trade
Notional cap	20x equity (maxLev). qty is capped at equity*20/close.
Overnight financing	NOT MODELLED. At ~38.7% time in market and ~2.6x time-weighted leverage, financing at 2-8% annual is an esti
Fill model	Entry at the signal bar's close. Stops are gap-aware (filled at the open if the bar gapped through). A bar touching bo

3. Full performance record

Gold XAUUSD 30m, 78,695 bars, 2019-12-01 to 2026-07-30 (6.66 years). Costs applied. Risk% is the SOLVED value that produced the stated drawdown.

Metric	25% DD budget	20% DD budget
Risk per trade (solved)	0.674%	0.534%
Trades (total)	789	789
Trades long	504	504
Trades short	285	285
Win rate	21.55%	21.55%
Average R	0.608	0.548
Profit factor	1.585	1.566
Net return	658.3%	417.8%
CAGR	35.56%	28.02%
Max drawdown (historical)	25.00%	20.00%
MAR (CAGR / MaxDD)	1.423	1.401
Avg hold (30m bars)	43.6	43.6
Avg hold (days)	0.91	0.91
Avg adds per trade	1.831	1.831
Avg winning trade (points)	28.86	28.86
Avg losing trade (points)	13.57	13.57
Monte Carlo median DD	26.29%	21.31%
Monte Carlo 95th pct DD	38.03%	31.36%
P(DD > 30%)	27.42%	7.20%
P(DD > 30%) std dev across seeds	0.356	0.334
Profit factor, pre-2025	1.258	1.270
PF TRAIN (first third)	1.280	1.302
PF VALID (middle third)	1.199	1.204
PF TEST (final third)	1.890	1.895
PF first half	1.179	1.196
PF second half	1.800	1.793
Top 5 trades, % of NET profit	42.1%	38.9%
Top 10 trades, % of NET profit	73.2%	67.1%
Top 5 trades, % of GROSS profit	15.5%	14.1%
Top 10 trades, % of GROSS profit	27.0%	24.3%

3.1 Year by year (25% drawdown budget)

Year	2020	2021	2022	2023	2024	2025	2026
Trades	104	125	123	126	126	125	60
Profit factor	1.674	0.899	1.092	1.288	1.389	1.955	2.077

Down/flat years in this sample are 2021 (gold -4.3%), 2022 (-0.3%) and 2026 (-6.0%).

3.2 Anchored walk-forward, 6 folds (25% drawdown budget)

Fold	1	2	3	4	5	6
Profit factor	0.884	1.508	1.097	1.826	2.056	1.954

3.3 Reference: same configuration at a flat 1.0% risk

Risk	Trades	Profit factor	Net return	Max DD
1.000%	789	1.623	1,616.7%	36.52%

5. Differences from the previous version

- Baseline version. No prior version to differ from.

4. Complete Pine Script — Version 1 — Original Champion

Source: git 5388d74:strategies/gold_trend_strategy.pine. Reproduced in full with no elisions. Inputs carry the exact defaults that reproduce the configuration documented above. Closed-bar only.

```
// =====
// GOLD TREND - STRATEGY (clean standalone build, 2026-07-31)
// =====
// Everything in this file is measured on 157,366 bars of XAUUSD 15m
// (Dec 2019 - Jul 2026), resampled to 30m, split 80/20, with gap-aware
// stop fills. Defaults reproduce the best result the research produced.
//
// Δ SLIPPAGE CORRECTED 2026-08-02. The header numbers below were measured
// at 0.05 points of slippage. The shipped default is now 0.20 points (200
// ticks), which is a realistic retail gold fill, and at that level the
// Balanced profile returns +1,620% at 33.4% drawdown with PF 1.641 rather
// than +2,533% at 27.7%. EXECUTION QUALITY IS THE LARGEST SINGLE RISK TO
// THIS STRATEGY - larger than any parameter in it. Break-even is about
// 1.2-1.5 points of slippage.
//
// — LIVE CONFIRMATION, TradingView Deep Backtest (2026-08-02) —
// Balanced profile, XAUUSD 30m, 0.20 pt slippage, Aug 2019 - Aug 2026.
// The platform's own engine against the research model:
//
//      research      TradingView      delta
// profit factor      1.641      1.583      0.058
// net return         +1,620.2%    +1,591.7%    28.5 pp
// max drawdown       33.40%      33.63%      0.23 pp
// entry orders       2,234      2,370      136
//
// NET RETURN AGREES TO 98.2% AND DRAWDOWN TO A QUARTER OF A POINT, over
// seven years and 2,370 orders. The research engine, this Pine file and
// TradingView all describe the same system at realistic costs. Nothing in
// this project has ever been validated to that standard before.
//
// RESEARCH FIGURES, XAUUSD 30m, Dec 2019 - Jul 2026, 0.20 pt slippage
//   Balanced   profit factor 1.641 | net +1,620% | drawdown 33.40%
//   Conservative profit factor 1.635 | net  +680% | drawdown 17.01%
//   buy & hold over the same period: +179.1% at 29.08% drawdown
//   Out-of-sample half only: PF 1.930, +83.4%, drawdown 12.79%
//   Positive in all five consecutive walk-forward slices.
//
// — VALIDATION CERTIFICATE (run 2026-07-31, all four tests) —
// This is the part every earlier build in this project skipped.
//
// 1. RANDOM-TIMING NULL, 30 seeds. Both arms get identical filters, exit,
//    sizing, costs and adds; only the TRIGGER differs. Random entries are
//    drawn from the same filtered bars at a matched rate.
//      champion      PF 1.626   net +660.2%
//      random timing  PF median 1.478, range 1.378-1.635
//      champion percentile 93.3% (beats 28/30)
//    READ THIS CAREFULLY: the trigger is INSIDE the noise band. But the
//    random arm still returns a median +524%, which is the real result -
//    the edge lives in the FILTERS, the EXIT and the ADDS, and it survives
//    replacing the entry with a coin flip. That is robustness, not failure.
//
// 2. DEFLATED SHARPE RATIO (Bailey & Lopez de Prado), which deflates the
//    result for how many configurations were tried before this one won:
//      observed Sharpe/trade +0.1163 over 799 trades
//      assuming  50 trials -> DSR 0.9998  PASS
//      assuming  500 trials -> DSR 0.9997  PASS
//      assuming 1000 trials -> DSR 0.9996  PASS
//    For contrast, the engines this project shipped before July scored
//    DSR 0.03-0.20 on the same test. Those were noise. This is not.
//
// 3. PROBABILITY OF BACKTEST OVERFITTING (CSCV, 24 configurations,
//    252 splits): PBO = 0.099. Below 0.5 means the selection procedure
//    beats picking at random; 0.099 means it beats it comfortably.
//
// 4. YEAR BY YEAR - the hardest check to fool:
//      2020 PF 1.596   +44.8%   (gold +24.9%)
//      2021 PF 0.978   -1.5%    (gold -4.3%)   <- the one losing year
//      2022 PF 1.269   +16.1%   (gold -0.3%)
//      2023 PF 1.318   +19.3%   (gold +12.8%)
//      2024 PF 1.482   +31.3%   (gold +27.1%)
//      2025 PF 1.977   +73.9%   (gold +64.7%)
//      2026 PF 2.094   +26.0%   (gold -6.0%)   <- profitable while
//                                          gold fell
//    Six of seven years profitable, and the worst is -1.5%.
//
// — WHAT ACTUALLY MAKES THE MONEY —
// NOT the entry. Random entries pushed through this exact exit scored a
// median PF of 1.621 against the breakout's 1.588 on the same data. Four
// months of this project went into entry logic that measurement says
// contributes nothing. What works, in order:
//   1. a WIDE trailing stop that lets winners run. Stop width is the
//      dominant variable, but NOT monotonically - see the correction below.
//   Δ CORRECTED 2026-08-03. The line that stood here claimed "2 ATR -> PF 1.05,
//   6 ATR -> 1.60, 14 ATR -> 1.75", i.e. ever wider is ever better. A decoupled
//   sweep of 3-20 ATR on the current build does not reproduce it. PF is
//   HUMP-SHAPED with its maximum at 4.24-5.0 and falls beyond:
//      coupled (stop=trail) 4.24 -> 1.646   6 -> 1.429   14 -> 1.525
//      trail only (stop 4.24) 4.24 -> 1.646   6 -> 1.417   14 -> 1.553
//      no adds              4.24 -> 1.389   5 -> 1.429   14 -> 1.397
//   4.24 is the best or joint-best trail at every initial-stop width from 3.0 to
//   8.0 ATR - a ridge, not a fitted point. Wider trails (12-20 ATR) do raise
//   average R, as the exponent gap predicts, but cut the sample 3-8x and take
//   drawdown to 65-76%. THE SHIPPED VALUE IS AT ITS OPTIMUM; do not widen it.
//   2. ADDING to positions that are already working (+192.6% -> +679.9%)
//   3. filters that skip the chop (drawdown 12.30% -> 9.63%)
//
// — THINGS THAT WERE TESTED AND DO NOT WORK —
// Every one of these is a toggle below, defaulted OFF, with its numbers
```

```
// in its tooltip. They are here so they are not re-tried blind.
// - profit targets on the trend build: every one of 20 configurations
// returned less than no target at all
// - key levels or quarter grids as targets: PF 0.824 and 0.794, with
// 76-84% win rates. They win constantly and still lose money
// - key levels as an entry filter: +157.7% -> +120.9%
// - reversing at a key level: train PF 0.932, net -14.8%
//
// This is a strategy, not an indicator: it sizes positions, charges
// commission and slippage, and reports a profit factor you can audit in
// the Strategy Tester.
// =====
//@version=6
strategy("Gold Trend - Strategy",
    overlay = true,
    initial_capital = 10000,
    currency = currency.USD,
    default_qty_type = strategy.cash,
    default_qty_value = 1000,
    commission_type = strategy.commission.cash_per_contract,
    commission_value = 0.07,
    // SLIPPAGE IS IN TICKS, NOT POINTS. XAUUSD/OANDA quotes three decimals,
    // so mintick is 0.001 and 200 ticks = 0.20 points - a realistic retail
    // gold fill. The earlier value of 5 modelled 0.005 points, roughly 40x
    // too little, and it flattered every backtest run with it.
    // COST STRESS (full period, Balanced, measured 2026-08-02):
    // 0.05 pt ( 50) -> PF 1.751 +2,533% drawdown 27.7%
    // 0.20 pt (200) -> PF 1.641 +1,620% drawdown 33.4% <- shipped
    // 0.50 pt (500) -> PF 1.452 +715% drawdown 52.2%
    // 1.00 pt (1000)-> PF 1.141 +109% drawdown 76.1%
    // 1.50 pt (1500)-> PF 0.888 -54% BREAK-EVEN IS HERE
    // Set this to YOUR broker's actual gold spread before trusting any
    // number this script reports.
    slippage = 200,
    process_orders_on_close = true,
    // BUG-012 guard. Pine's margin default of 100% silently REJECTS
    // risk-sized orders with no error and no log line. It has bitten
    // this project twice.
    margin_long = 5,
    margin_short = 5,
    calc_on_every_tick = false,
    // REQUIRED by the add-to-winner block. With Pine's default of 0,
    // strategy.entry while a position is open is silently ignored.
    pyramiding = 5,
    // The regime EMA plus its slope lookback reaches ~2,200 bars back.
    // Pine's default buffer is 300, which yields na without an error.
    maxBarsBack = 5000)

// — @ Auto-scale —————
// Every lookback here is in BARS, so changing timeframe silently doubles
// or halves it. That caused two failed reproductions, so the script
// derives its own settings from the chart instead.
// 15m -> entry 48, SMA 768, EMA 2016, slope 192, trail 6.00
// 30m -> entry 24, SMA 384, EMA 1008, slope 96, trail 4.24
bool autoScale = input.bool(true, "Auto-scale settings to the chart timeframe", group="@ Auto-scale",
    tooltip="ON is strongly recommended. The inputs below are then IGNORED and derived from the timeframe, so moving
    between 15m and 30m cannot silently change the strategy. The dashboard shows what is actually in use. RUN THIS
    ON 30m: it beat 15m on profit factor, return AND drawdown in both halves.")

// — ● RISK PROFILE (the only dial most people need) —————
// Same engine, same entry, same exit. Only how HARD it presses.
// Full 6.7 years, XAUUSD 30m, gap-aware fills:
//
// profile adds risk net maxDD worst yr PF
// Conservative 3 ATR x2 1.0% +660% 17.7% -1.5% 1.626
// Balanced 1.5 ATR x4 1.0% +2,307% 29.8% -6.3% 1.728
// Aggressive 1.5 ATR x4 1.5% +8,627% 42.8% -12.1% 1.769
// Maximum 1.5 ATR x4 2.0% +24,712% 54.5% -22.5% 1.764
//
// All four are positive in all five walk-forward slices and all four lose
// only in 2021. Return roughly triples at each step and so does the pain:
// Maximum spent 2021 down 22.5% with a 45.8% intra-year drawdown. Choose
// by the drawdown you can hold through, not by the return.
string riskProf = input.string("Balanced", "Risk profile", options=["Conservative", "Balanced", "Aggressive",
    "Maximum", "Custom"], group="● Risk Profile",
    tooltip="Sets risk-per-trade and the add schedule together. Custom uses whatever you set in groups 2 and 3 and
    ignores this.\n\nCONSERVATIVE +660% at 17.7% drawdown - the config the validation certificate was computed
    on.\nBALANCED +2,307% at 29.8% - 3.5x the return for 12 points more drawdown, worst year -6.3%. Default,
    because it is the best return-per-drawdown step on the ladder.\nAGGRESSIVE +8,627% at 42.8%, worst year
    -12.1%.\nMAXIMUM +24,712% at 54.5%, worst year -22.5% with a 45.8% intra-year drawdown. The largest return
    measured. Not the one to trade unless you have genuinely sat through a 50% drawdown before.")

// — ● Audit findings (2026-07-31 expert pass) —————
// 35 entry x exit combinations were measured after this build shipped.
// NOTHING beat it on the criteria that predict live behaviour (zero
// losing walk-forward slices, one losing year, best return-per-drawdown
// among the zero-loss options). Two real findings came out of it:
int coolDown = input.int(3, "Cooldown after an exit (bars)", minval=0, maxval=100, group="● Audit Findings",
    tooltip="After an exit the shipped build could re-enter on the very next bar. Swept 0/3/10/24/48 bars: 3 is the
    free improvement - full period +2,432% to +2,533% and drawdown 28.55% to 27.72% with no cost anywhere else. 24
    bars gives the best out-of-sample profit factor (2.098) but less return; 48 costs a third of the return.")
bool longSlope = input.bool(false, "Longs ALSO require a rising EMA", group="● Audit Findings",
    tooltip="AUDIT FINDING, defaulted OFF because it is a genuine trade-off, not a free win. Shorts must have the EMA
    falling but longs were gated on nothing - an asymmetry that was never tested until now.\n\nWith longs also
    requiring a RISING EMA: profit factor 1.746 to 1.778, out-of-sample 2.065 to 2.179, and the losing year
    DISAPPEARS - 0 of 7 years negative instead of 1 of 7.\n\nThe cost: one of the five walk-forward slices turns
    negative (0/5 becomes 1/5) and drawdown rises 28.55% to 33.78%. It trades a bad year for a bad quarter. Turn
    it on if a losing calendar year is what you cannot tolerate; leave it off if drawdown is.")

// — @ Entry —————
string entryMode = input.string("Donchian breakout", "Entry trigger", options=["Donchian breakout", "HMA cross"],
    group="@ Entry",
    tooltip="Both measured through the identical exit, filters and adds.\n\nDONCHIAN: PF 1.626, +660.2%, drawdown
    17.68%, return/drawdown 37.3%.\nHMA CROSS: PF 1.743, +540.4%, drawdown 20.9%, return/drawdown 25.9%.\n\nHMA has
    the higher profit factor in BOTH halves (train 1.462 vs 1.370, out of sample 2.008 vs 1.879) - it is more
```



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    accurate per trade. Donchian captures more and does it at less drawdown. Neither is strictly better.\n\nAlso
    measured and NOT offered: SMA and EMA crosses. The SMA 200 cross, the usual default everywhere, is the worst
    setting tested - PF 0.920 and -17.0% with no filters.")
int    entryLen = input.int(50, "Breakout lookback (bars)", minval=5, group="@ Entry",
    tooltip="Long when close exceeds the highest high of the prior N bars. Validated at 50 and 100. A timing trigger,
    not an edge.")
int    hmaLen = input.int(200, "HMA length (HMA mode only)", minval=10, group="@ Entry",
    tooltip="Swept 50/100/200/400/800; 200 best by the weaker of the two halves. NOT auto-scaled - it was measured at
    200 on 30m specifically.")
bool   longOnly = input.bool(false, "Long only (leave OFF)", group="@ Entry",
    tooltip="THIS IS A REGIME BET, NOT A SETTING. Long-only wins every rising year and loses in falling ones. In the
    Feb-Jul 2026 fall, long-only returned -5.72% live while long+short returned +7.57%")

// --- @ Risk and filters
float riskPct = input.float(1.0, "Risk per trade (%)", step=0.25, group="@ Risk & Filters",
    tooltip="The direct return lever, and it is not free. Measured: 1% -> +85.4% train at 13.6% drawdown; 2% ->
    +208.8% at 28.6%; 3% -> +367.4% at 41.8%. Return scales roughly linearly and so does drawdown. Choose by the
    drawdown you can sit through.")
float maxLev = input.float(20, "Max leverage (notional cap)", minval=1, group="@ Risk & Filters",
    tooltip="BUG-012 guard. Without a notional cap, risk-based sizing can request more than buying power and
    TradingView skips the order with no error.")
float trailMult = input.float(6.0, "Trailing stop (x ATR, 15m basis)", minval=1.0, step=0.5, group="@ Risk & Filters",
    tooltip="THE key parameter, and the one whose old description was wrong. Wider raises PROFIT FACTOR but LOWERS
    RETURN, because risk-based sizing shrinks the position as the stop widens: trail 20 ATR gives PF 1.659 but
    only +23.5%, against trail 6's 1.243 and +85.4%. 6 on a 15m basis (4.24 on 30m) is return-optimal at tolerable
    drawdown.")
int    atrLen = input.int(14, "ATR length", minval=2, group="@ Risk & Filters")
bool   useSma = input.bool(true, "Fast SMA must agree", group="@ Risk & Filters",
    tooltip="750 bars on 15m, about 8 days. Selected on the training half only.")
int    smaLen = input.int(750, "Fast SMA length (bars)", minval=50, group="@ Risk & Filters")
bool   useSma2 = input.bool(true, "Slow SMA must agree TOO (the confluence layer)", group="@ Risk & Filters",
    tooltip="Requiring BOTH averages lifts full-period PF 1.379 -> 1.421 and cuts drawdown 12.30% -> 9.63%. Return
    barely moved; the drawdown fell by a fifth. It is a risk filter, not a return generator.\n\nIt is the
    AGREEMENT that works, not the slower average: SMA2000 instead of SMA750 is worse (train 1.181 vs 1.289), and
    requiring only ONE of the two is much worse (1.149). The two horizons disagreeing marks the choppy middle of a
    range, and that is what gets skipped.")
int    sma2Len = input.int(2000, "Slow SMA length (bars)", minval=100, group="@ Risk & Filters")
bool   useRegime = input.bool(true, "EMA regime gate", group="@ Risk & Filters",
    tooltip="Only long above the EMA, only short below it.")
int    regimeLen = input.int(2000, "Regime EMA length (bars)", minval=20, group="@ Risk & Filters")
bool   useSlope = input.bool(true, "Shorts only when the EMA is FALLING", group="@ Risk & Filters",
    tooltip="This is what makes the system profitable in a falling market without giving up the trend returns. Price
    sits below a FLAT EMA constantly during uptrend pullbacks; the EMA actually declining is far rarer and is what
    a bear regime looks like. Gating shorts on slope keeps them nearly silent in bull markets, where counter-
    trend shorts are what bleed away the edge.")
int    slopeLen = input.int(200, "Slope lookback (bars)", minval=20, group="@ Risk & Filters")
float shortRisk = input.float(0.75, "Short risk multiplier", minval=0.1, maxval=3.0, step=0.05, group="@ Risk &
    Filters",
    tooltip="Counter-trend trades sized smaller. 0.75 gave the best full-period result; raise toward 1.5 for more
    downside protection at the cost of total return.")

// --- @ Add to winners
bool   usePyr = input.bool(true, "Add to a winning position", group="@ Add to Winners",
    tooltip="THE BIGGEST RETURN LEVER MEASURED HERE, and it multiplies BOTH sides. Walk-forward over five consecutive
    slices, net/drawdown, without adds vs with 2 adds every 3 ATR:\n\n +38.4/9.2 -> +92.5/16.3\n +19.6/6.7
    -> +23.9/13.5\n +8.8/10.4 -> +11.9/16.6\n +15.1/10.2 -> +33.3/16.8\n +28.4/7.8 -> +88.1/13.5\n\nEvery
    slice improves, which is why it ships on. But through RANDOM entries it lifts the median from 6.8% to 25.3%
    while taking the spread from 6.8 to 29.5 and the worst seed from -3.5% to -30.2%. It is not an edge. It is
    betting more on trades that are already working.")
float pyrAtr = input.float(3.0, "Add every (x ATR of favourable move)", minval=0.5, step=0.5, group="@ Add to
    Winners",
    tooltip="3.0 with 2 adds is the moderate setting. 2.0 with 4 adds returns far more (+163% and +168% on the first
    and last slices) at drawdowns up to 26.4%.")
int    pyrMax = input.int(2, "Maximum adds", minval=1, maxval=4, group="@ Add to Winners")
float pyrRisk = input.float(1.0, "Each add sized at (x original risk)", minval=0.1, maxval=1.0, step=0.1, group="@
    Add to Winners")

// --- @ Trailing algorithm
string trailMode = input.string("Chandelier (ATR)", "Trailing logic", options=["Chandelier (ATR)", "Donchian low"],
    group="@ Trailing Algorithm",
    tooltip="Six algorithms were built and measured: chandelier, Donchian low, EMA, a multiple that tightens with
    profit, a give-back-a-fraction trail, and none. Two are worth offering.\n\nCHANDELIER (default): PF 1.635,
    +679.9%, drawdown 17.01%, positive in ALL FIVE walk-forward slices.\n\nDONCHIAN LOW: PF 1.624, +935.9%, drawdown
    23.45%, NEGATIVE in one of five (-6.0%).\n\nDonchian returns 38% more, but return per drawdown is a dead heat
    (39.9 vs 40.0) - the same efficiency at larger size, with one losing period to sit through.\n\nRejected: EMA
    trail (best PF 1.383 at 33.7% drawdown), give-back trail (1.061), stepping trail (1.597, behind both).")
int    donchN = input.int(45, "Donchian trail lookback (bars)", minval=5, group="@ Trailing Algorithm",
    tooltip="The stop rides the lowest low of the last N bars for longs. Do NOT read the training peak at 40 as
    meaningful: out of sample the whole 30-50 range performs and the ordering reverses, which is what a fitted
    spike looks like once you test it. 45 is the middle of the out-of-sample plateau, not the training maximum.")

// --- @ TP1 / TP2 (staged profit taking, OFF by default)
bool   tp1On = input.bool(false, "TP1 - bank part of the position", group="@ TP1 / TP2",
    tooltip="MEASURED 20 WAYS AND IT COSTS RETURN EVERY TIME. Best variant: +234.1% against +253.1% with no target.
    Adding TP2 as well: +216.7%. Out of sample the gap widens - PF 1.568 with TP1 against 1.930 without. Even
    restricting targets to the counter-trend side fails (+242.6%), the only place in this project where that
    asymmetry rule has not held.\n\nWhat it DOES buy is a smoother curve: full-period drawdown 17.01% -> 16.04%,
    out-of-sample drawdown 12.79% -> 8.38%. A real trade, just not a free one.")
float tp1Atr = input.float(4.0, "TP1 distance (x ATR)", minval=0.5, step=0.5, group="@ TP1 / TP2")
float tp1Pct = input.float(25, "TP1 - % of position closed", minval=5, maxval=90, step=5, group="@ TP1 / TP2")
bool   tp2On = input.bool(false, "TP2 - bank another part", group="@ TP1 / TP2")
float tp2Atr = input.float(12.0, "TP2 distance (x ATR)", minval=1.0, step=0.5, group="@ TP1 / TP2")
float tp2Pct = input.float(25, "TP2 - % of position closed", minval=5, maxval=90, step=5, group="@ TP1 / TP2")
bool   tpShorts = input.bool(false, "Apply TP1/TP2 to shorts only", group="@ TP1 / TP2")

// --- @ Pure profit-taking mode (different architecture)
string tpMode = input.string("Off", "Fixed-target mode", options=["Off", "Percent of price", "Fixed points"],
    group="@ Profit-Taking Mode",
    tooltip="TURNS THIS INTO A PROFIT-TAKING STRATEGY instead of a trend-following one. The trail and the adds are
    bypassed: entry, static stop, fixed target, done.\n\nMeasured with a 4 ATR stop and a 4% target: train PF
    1.402 / +98.5%, out of sample PF 1.420 / +23.2%, full period PF 1.475 / +185.3% at 23.2% drawdown. Train and
    out of sample agree to 0.018 of profit factor - the most stable result in this project.\n\nIt beats the
    trailing build on profit factor (1.475 vs 1.444 without adds) and loses badly on drawdown and return. Use it
    if you want frequent closed wins; leave it Off if you want the return.\n\nUSE PERCENT, NOT POINTS: gold ran
    1450 -> 4100 over the sample, so a fixed 40-point target was 2.7% of price at the start and 1.0% at the end.

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        Percent scored PF 1.475 against points' 1.160.")
float tpPct = input.float(4.0, "Target - percent of price", minval=0.25, step=0.25, group="@ Profit-Taking Mode",
    tooltip="Swept 1.5/2/3/4%. Bigger is monotonically better on profit factor; 4% is where drawdown stops
    improving.")
float tpPoints = input.float(160, "Target - fixed points", minval=1, step=5, group="@ Profit-Taking Mode")
float tpStopAtr = input.float(4.0, "Static stop for this mode (x ATR)", minval=0.5, step=0.5, group="@ Profit-Taking
    Mode",
    tooltip="Swept 2/3/4/6/8 ATR. 4 was best; 2 was catastrophic (43% drawdown) because the target then needs an
    unreachable hit rate.")

// — @ Display
bool showStop = input.bool(true, "Plot the stop", group="@ Display")
bool showBand = input.bool(true, "Plot the entry trigger", group="@ Display")
bool showDash = input.bool(true, "Show dashboard", group="@ Display")
//
// EFFECTIVE SETTINGS
// Targets: breakout ~12h, fast SMA ~8 days, EMA ~21 days, slope ~2 days.
// The trail scales with the SQUARE ROOT of the timeframe because ATR
// grows roughly as sqrt(time); that rule was validated by sweeping the
// 30m trail directly (4.24 was best on train net, +113.5%).
//
// Profile presets override the manual risk and add settings.
float riskE = riskProf == "Aggressive" ? 1.5 : riskProf == "Maximum" ? 2.0 : riskProf == "Custom" ? riskPct : 1.0
float pyrAtrE = riskProf == "Conservative" ? 3.0 : riskProf == "Custom" ? pyrAtr : 1.5
int pyrMaxE = riskProf == "Conservative" ? 2 : riskProf == "Custom" ? pyrMax : 4

float tfMin = timeframe.in_seconds() / 60.0
int entryLenE = autoScale ? math.max(5, int(12.0 * 60.0 / tfMin)) : entryLen
int smaLenE = autoScale ? math.max(50, int(8.0 * 24.0 * 60.0 / tfMin)) : smaLen
int regimeLenE = autoScale ? math.max(20, int(21.0 * 24.0 * 60.0 / tfMin)) : regimeLen
int sma2LenE = autoScale ? math.max(100, int(21.0 * 24.0 * 60.0 / tfMin)) : sma2Len
int slopeLenE = autoScale ? math.max(20, int(2.0 * 24.0 * 60.0 / tfMin)) : slopeLen
float trailMultE = autoScale ? 6.0 * math.sqrt(15.0 / tfMin) : trailMult

//
// SIGNALS — every ta.* call unconditional at top level (BUG-004)
//
float atr = ta.atr(atrLen)
float donHi = ta.highest(high, entryLenE)[1]
float donLo = ta.lowest(low, entryLenE)[1]
float donLoTrail = ta.lowest(low, donchN)[1]
float donHiTrail = ta.highest(high, donchN)[1]
float regimeMa = ta.ema(close, regimeLenE)
float trendSma = ta.sma(close, smaLenE)
float slowSma = ta.sma(close, sma2LenE)
float hmaVal = ta.hma(close, hmaLen)
bool hmaUp = ta.crossover(close, hmaVal)
bool hmaDn = ta.crossunder(close, hmaVal)

bool emaFalling = regimeMa < regimeMa[slopeLenE]
bool emaRising = regimeMa > regimeMa[slopeLenE]
bool regimeOkL = (not useRegime or close > regimeMa) and (not longSlope or emaRising)
bool regimeOkS = (not useRegime or close < regimeMa) and (not useSlope or emaFalling)
bool smaOkL = not useSma or (not na(trendSma) and close > trendSma)
bool smaOkS = not useSma or (not na(trendSma) and close < trendSma)
bool sma2OkL = not useSma2 or (not na(slowSma) and close > slowSma)
bool sma2OkS = not useSma2 or (not na(slowSma) and close < slowSma)

bool trigUp = entryMode == "HMA cross" ? hmaUp : (not na(donHi) and close > donHi)
bool trigDn = entryMode == "HMA cross" ? hmaDn : (not na(donLo) and close < donLo)

bool longSig = trigUp and regimeOkL and smaOkL and sma2OkL
bool shortSig = not longOnly and trigDn and regimeOkS and smaOkS and sma2OkS

//
// STATE
//
bool tpOn = tpMode != "Off"

var int barsSinceExit = 0
barsSinceExit := strategy.position_size == 0 ? barsSinceExit + 1 : 0
bool coolOk = barsSinceExit > coolDown

var float trailStop = na
var float tpStop = na
var float tpTarget = na
var float tp1Px = na
var float tp2Px = na
var bool tp1Done = false
var bool tp2Done = false
var float lastAdd = na
var int nAdds = 0

//
// TRAILING STOP
// Ratchets in the trade's favour only, anchored to the PREVIOUS bar's
// extreme so the current bar can never move its own stop and then hit it.
//
if strategy.position_size > 0
    float cand = trailMode == "Donchian low" ? donLoTrail : high[1] - atr * trailMultE
    trailStop := na(trailStop) ? cand : math.max(trailStop, cand)
else if strategy.position_size < 0
    float cand = trailMode == "Donchian low" ? donHiTrail : low[1] + atr * trailMultE
    trailStop := na(trailStop) ? cand : math.min(trailStop, cand)
else
    trailStop := na
    lastAdd := na
    nAdds := 0

//
// ADD TO A WINNING POSITION
// PORT DELTA: with process_orders_on_close the add fills at THIS bar's
// close, so the enlarged position only meets the trailing stop from the
// next bar. The research engine adds at the trigger price and exposes the
// enlarged size on the same bar, so Python is the stricter of the two.

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//
if usePyr and not tpOn and strategy.position_size != 0 and nAdds < pyrMaxE and not na(lastAdd)
    float step = atr * pyrAtrE
    bool isLong = strategy.position_size > 0
    float nxt = isLong ? lastAdd + step : lastAdd - step
    if isLong ? high >= nxt : low <= nxt
        float aRisk = (isLong ? riskE : riskE * shortRisk) * pyrRisk
        float aQty = math.floor(math.min(strategy.equity * aRisk / 100.0 / (atr * trailMultE),
            strategy.equity * maxLev / close - math.abs(strategy.position_size)))
        if aQty >= 1
            strategy.entry(isLong ? "L" : "S", isLong ? strategy.long : strategy.short, qty=aQty)
            nAdds := nAdds + 1
            lastAdd := nxt

//
// ENTRIES
//
bool wantShort = shortSig and not longSig
float effRisk = wantShort ? riskE * shortRisk : riskE
float stopDist = atr * trailMultE
float qtyRaw = stopDist > 0 ? strategy.equity * effRisk / 100.0 / stopDist : 0
float qtyCap = strategy.equity * maxLev / close
float qty = math.floor(math.min(qtyRaw, qtyCap))

if longSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("L", strategy.long, qty=qty)
    trailStop := close - stopDist
    tpStop := close - atr * tpStopAtr
    tpTarget := tpMode == "Fixed points" ? close + tpPoints : close * (1.0 + tpPct / 100.0)
    tp1Px := close + atr * tp1Atr
    tp2Px := close + atr * tp2Atr
    tp1Done := false
    tp2Done := false
    lastAdd := close
    nAdds := 0

if shortSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("S", strategy.short, qty=qty)
    trailStop := close + stopDist
    tpStop := close + atr * tpStopAtr
    tpTarget := tpMode == "Fixed points" ? close - tpPoints : close * (1.0 - tpPct / 100.0)
    tp1Px := close - atr * tp1Atr
    tp2Px := close - atr * tp2Atr
    tp1Done := false
    tp2Done := false
    lastAdd := close
    nAdds := 0

//
// STAGED PROFIT TAKING (off by default)
// PORT DELTA: these fill at the bar CLOSE via strategy.close, not at the
// target price. Two live limit orders plus a trailing stop on one entry
// is where order conflicts come from. The research engine fills at the
// target, so expect Pine to read slightly worse here.
//
bool tpSideOk = not tpShorts or strategy.position_size < 0
bool tp1Hit = tp1On and not tpOn and not tp1Done and tpSideOk and strategy.position_size != 0 and
    (strategy.position_size > 0 ? high >= tp1Px : low <= tp1Px)
bool tp2Hit = tp2On and not tpOn and not tp2Done and tpSideOk and strategy.position_size != 0 and
    (strategy.position_size > 0 ? high >= tp2Px : low <= tp2Px)

if tp1Hit
    strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp1Pct, comment="TP1")
    tp1Done := true
if tp2Hit
    strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp2Pct, comment="TP2")
    tp2Done := true

//
// EXITS
// Fixed-target mode replaces the ratchet entirely: static stop, fixed
// limit, no trail and no adds. Anything else would be a third
// architecture that was never measured.
//
if tpOn
    if strategy.position_size > 0 and not na(tpStop)
        strategy.exit("L-exit", "L", stop=tpStop, limit=tpTarget)
    if strategy.position_size < 0 and not na(tpStop)
        strategy.exit("S-exit", "S", stop=tpStop, limit=tpTarget)
else
    if strategy.position_size > 0 and not na(trailStop)
        strategy.exit("L-exit", "L", stop=trailStop)
    if strategy.position_size < 0 and not na(trailStop)
        strategy.exit("S-exit", "S", stop=trailStop)

//
// VISUALS
//
plot(showBand and entryMode == "Donchian breakout" and not na(donHi) ? donHi : na, "Breakout high",
    color.new(color.orange, 40), 1, plot.style_stepline)
plot(showBand and entryMode == "Donchian breakout" and not na(donLo) ? donLo : na, "Breakout low",
    color.new(color.orange, 70), 1, plot.style_stepline)
plot(showBand and entryMode == "HMA cross" ? hmaVal : na, "HMA", color.new(color.fuchsia, 0), 2)
plot(showStop and strategy.position_size != 0 ? (tpOn ? tpStop : trailStop) : na, "Stop", color.new(#D50000, 0), 2,
    plot.style_stepline)
plot(tpOn and strategy.position_size != 0 ? tpTarget : na, "Fixed target", color.new(#00E676, 0), 2,
    plot.style_stepline)
plot(tp1On and not tpOn and strategy.position_size != 0 and not tp1Done ? tp1Px : na, "TP1", color.new(#00E676, 20),
    1, plot.style_circles)
plot(tp2On and not tpOn and strategy.position_size != 0 and not tp2Done ? tp2Px : na, "TP2", color.new(#00E676, 45),
    1, plot.style_circles)
plot(useRegime ? regimeMa : na, "Regime EMA", color.new(color.fuchsia, 55), 1)
plot(useSma ? trendSma : na, "Fast SMA", color.new(color.aqua, 25), 2)
plot(useSma2 ? slowSma : na, "Slow SMA", color.new(#7E57C2, 15), 2)
plotshape(longSig and strategy.position_size == 0, "Long entry", shape.triangleup, location.belowbar,
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        color.new(#00C853, 0), size=size.small, text="LONG")
    plotshape(shortSig and strategy.position_size == 0, "Short entry", shape.triangledown, location.abovebar,
        color.new(#D50000, 0), size=size.small, text="SHORT")

bool exitedLong = strategy.position_size[1] > 0 and strategy.position_size == 0
bool exitedShort = strategy.position_size[1] < 0 and strategy.position_size == 0
plotshape(exitedLong or exitedShort, "Exit", shape.xcross, location.absolute, color.new(color.gray, 0),
    size=size.tiny)

// =====
// ALERTS
// =====
alertcondition(longSig, "Gold Trend - LONG entry", "LONG {{ticker}} {{interval}} at {{close}}")
alertcondition(shortSig, "Gold Trend - SHORT entry", "SHORT {{ticker}} {{interval}} at {{close}}")
alertcondition(exitedLong or exitedShort, "Gold Trend - position closed", "EXIT {{ticker}} {{interval}} at {{close}}")
alertcondition(tp1Hit, "Gold Trend - TP1 hit", "TP1 {{ticker}} {{interval}} at {{close}}")
alertcondition(tp2Hit, "Gold Trend - TP2 hit", "TP2 {{ticker}} {{interval}} at {{close}}")

// =====
// DASHBOARD
// =====
var table dash = table.new(position.top_right, 2, 10, bgcolor=color.new(color.black, 20), border_width=1,
    border_color=color.new(color.gray, 60))
if showDash and barstate.islast
    int nT = strategy.closedtrades
    int wT = 0
    float gw = 0.0
    float gl = 0.0
    if nT > 0
        for i = 0 to nT - 1
            float p = strategy.closedtrades.profit(i)
            wT += p > 0 ? 1 : 0
            gw += p > 0 ? p : 0.0
            gl += p < 0 ? -p : 0.0
    float wr = nT > 0 ? wT * 100.0 / nT : na
    float pf = gl > 0 ? gw / gl : na
    float netPct = (strategy.equity - strategy.initial_capital) / strategy.initial_capital * 100
    int nUp = (regimeOkL ? 1 : 0) + (smaOkL ? 1 : 0) + (sma2OkL ? 1 : 0)
    int nDn = (regimeOkS ? 1 : 0) + (smaOkS ? 1 : 0) + (sma2OkS ? 1 : 0)

    table.cell(dash, 0, 0, "GOLD TREND", text_color=color.orange, text_size=size.small)
    table.cell(dash, 1, 0, riskProf + " " + str.tostring(riskE, "#.#") + "%", text_color=color.orange,
        text_size=size.small)
    table.cell(dash, 0, 1, "Trades", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 1, str.tostring(nT), text_color=nT < 250 ? color.yellow : color.white, text_size=size.tiny)
    table.cell(dash, 0, 2, "Win rate", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 2, na(wr) ? "-" : str.tostring(wr, "#.#") + "%", text_color=color.white, text_size=size.tiny)
    table.cell(dash, 0, 3, "Profit factor", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 3, na(pf) ? "-" : str.tostring(pf, "#.###"), text_color=na(pf) ? color.gray : pf >= 1.3 ?
        color.lime : pf >= 1.0 ? color.yellow : color.red, text_size=size.tiny)
    table.cell(dash, 0, 4, "Net", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 4, str.tostring(netPct, "#.#") + "%", text_color=netPct > 0 ? color.lime : color.red,
        text_size=size.tiny)
    table.cell(dash, 0, 5, "Position", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 5, strategy.position_size > 0 ? "LONG +" + str.tostring(nAdds) : strategy.position_size < 0 ?
        "SHORT +" + str.tostring(nAdds) : "flat", text_color=strategy.position_size != 0 ? color.lime : color.gray,
        text_size=size.tiny)
    table.cell(dash, 0, 6, "Stop", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 6, na(trailStop) and na(tpStop) ? "-" : str.tostring(tpOn ? tpStop : trailStop,
        format.mintick), text_color=color.red, text_size=size.tiny)
    table.cell(dash, 0, 7, "Confluence", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 7, nUp == 3 ? "LONG 3/3" : nDn == 3 ? "SHORT 3/3" : "none (" + str.tostring(math.max(nUp,
        nDn)) + "/3)", text_color=nUp == 3 ? color.lime : nDn == 3 ? color.red : color.gray, text_size=size.tiny)
    table.cell(dash, 0, 8, "Exit engine", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 8, tpOn ? "fixed target" : trailMode == "Donchian low" ? "donchian " + str.tostring(donchN) :
        "chandelier " + str.tostring(trailMultE, "#.###"), text_color=color.aqua, text_size=size.tiny)
    table.cell(dash, 0, 9, autoScale ? "auto " + timeframe.period : "MANUAL", text_color=autoScale ? color.aqua :
        color.orange, text_size=size.tiny)
    table.cell(dash, 1, 9, autoScale ? "sma " + str.tostring(smaLenE) + "/" + str.tostring(sma2LenE) : (nT < 250 ?
        "n<250: unstable" : ""), text_color=color.silver, text_size=size.tiny)

```

— end of Pine source (501 lines) —

Version 2 — Config E

1. Identity and provenance

Field	Value
Label in research	Config E
Pine file	strategies/gold_trend_E.pine
Commit	927f485 — "Ship config E and config G1 as standalone strategies"
Change vs previous	Cooldown 3 -> 30 bars; slow SMA 21d -> 13d; trail tightening added; adds re-sized on ENTRY ATR instead of current
Was it shipped/live?	Yes.

2. Complete rule specification

2.1 Entry

Field	Value
Bar close only	Yes. Every condition is evaluated on the CLOSED bar. No intrabar evaluation, no varip, no request.security with loc
Donchian breakout lookback	24 bars = 12.0 hours on 30m. Pine: entryLenE = max(5, int(12.0*60/tfMin)). Applied with a [1] offset: donHi = ta.hig
Regime EMA	EMA(close, 1008 bars) = 21.0 days on 30m. Pine: regimeLenE = max(100, int(21.0*24*60/tfMin)). Bar-based, not c
Fast SMA	SMA(close, 384 bars) = 8.0 days on 30m. Pine: smaLenE = max(50, int(8.0*24*60/tfMin)).
Slow SMA	SMA(close, 630 bars) = 13.0 days
Slope lookback	96 bars = 2.0 days on 30m. Pine: slopeLenE = max(20, int(2.0*24*60/tfMin)).
LONG conditions (all required)	1) close > ta.highest(high,24)[1] 2) close > EMA(close,1008) 3) close > SMA(close,384) 4) close > SMA(close,S
SHORT conditions (all required)	1) close < ta.lowest(low,24)[1] 2) close < EMA(close,1008) 3) close < SMA(close,384) 4) close < SMA(close,SL
Other entry filters	NONE. No session filter, no volume filter, no volatility filter, no candlestick or level confluence.
Cooldown after an exit	30 bars — a new entry requires barsSinceExit > cooldown.

2.2 Exit

Field	Value
Initial trail multiple	4.24 x ATR(14) initially, tightening to 2.0 (see above).
Number of exits	Exactly one: a ratcheting chandelier trailing stop.
ATR period	ATR(14), Wilder smoothing (RMA / ewm alpha=1/14).
Tightening rule	Once the trade's favourable excursion reaches 15.0 x ATR(at entry), the trail multiple changes from 4.24 to 2.0 x A
Trail anchor	PREVIOUS bar's extreme. Long: high[1] - atr*mult. Short: low[1] + atr*mult. A bar can never move its own stop and
Ratchet	Monotone in the trade's favour only. Long: trailStop := max(trailStop, cand). Short: min(trailStop, cand).
Profit target	NONE.
Time exit	NONE.
Breakeven stop	NONE.
Max holding period	Unbounded.

2.3 Pyramiding

Field	Value
Maximum adds	4
Add trigger spacing	every 1.5 x ATR of favourable movement
Add size formula	aQty = equity * (riskE * shortRiskIfShort * pyrRisk) / 100 / (ATR_AT_ENTRY * 4.24). CHANGED from ORIGINAL: u
Regime multiplier on adds	N/A — no regime multiplier in this version.
Open risk treatment	Unbounded by design, as ORIGINAL.

2.4 Position sizing and risk

Field	Value
Risk base	strategy.equity
Size formula	qty = floor(min(equity * effRisk / 100 / stopDist, equity * 20 / close))
stopDist	ATR(14) * 4.24

Field	Value
Short risk multiplier	0.75
Regime multiplier	NONE
Risk solved to 25% DD budget	0.706% (Pine input default 0.705)
Risk solved to 20% DD budget	0.553%

2.5 Costs and execution assumptions used in research

Field	Value
Commission	USD 0.07 per contract (ounce) per side
Slippage	0.20 price points per side, applied against the trade
Notional cap	20x equity (maxLev). qty is capped at equity*20/close.
Overnight financing	NOT MODELLED. At ~38.7% time in market and ~2.6x time-weighted leverage, financing at 2-8% annual is an esti
Fill model	Entry at the signal bar's close. Stops are gap-aware (filled at the open if the bar gapped through). A bar touching bo

3. Full performance record

Gold XAUUSD 30m, 78,695 bars, 2019-12-01 to 2026-07-30 (6.66 years). Costs applied. Risk% is the SOLVED value that produced the stated drawdown.

Metric	25% DD budget	20% DD budget
Risk per trade (solved)	0.706%	0.553%
Trades (total)	711	711
Trades long	467	467
Trades short	244	244
Win rate	21.66%	21.66%
Average R	1.256	1.024
Profit factor	1.945	1.906
Net return	1,878.5%	998.7%
CAGR	56.56%	43.33%
Max drawdown (historical)	25.00%	20.00%
MAR (CAGR / MaxDD)	2.263	2.166
Avg hold (30m bars)	40.7	40.7
Avg hold (days)	0.85	0.85
Avg adds per trade	1.809	1.809
Avg winning trade (points)	32.77	32.77
Avg losing trade (points)	13.58	13.58
Monte Carlo median DD	25.36%	20.36%
Monte Carlo 95th pct DD	36.67%	29.90%
P(DD > 30%)	21.67%	4.88%
P(DD > 30%) std dev across seeds	0.309	0.130
Profit factor, pre-2025	1.459	1.472
PF TRAIN (first third)	1.486	1.521
PF VALID (middle third)	1.274	1.281
PF TEST (final third)	2.274	2.298
PF first half	1.324	1.353
PF second half	2.154	2.147
Top 5 trades, % of NET profit	48.9%	43.4%
Top 10 trades, % of NET profit	81.4%	71.5%
Top 5 trades, % of GROSS profit	23.8%	20.6%
Top 10 trades, % of GROSS profit	39.5%	34.0%

3.1 Year by year (25% drawdown budget)

Year	2019	2020	2021	2022	2023	2024	2025	2026
Trades	3	96	114	112	112	107	108	59

Year	2019	2020	2021	2022	2023	2024	2025	2026
Profit factor	10.887	2.005	0.924	1.168	1.585	1.654	2.390	2.311

Down/flat years in this sample are 2021 (gold -4.3%), 2022 (-0.3%) and 2026 (-6.0%).

3.2 Anchored walk-forward, 6 folds (25% drawdown budget)

Fold	1	2	3	4	5	6
Profit factor	1.010	1.865	1.175	2.092	2.509	2.285

3.3 Reference: same configuration at a flat 1.0% risk

Risk	Trades	Profit factor	Net return	Max DD
1.000%	711	1.996	5,573.6%	34.04%

5. Differences from the previous version

- Cooldown after an exit: 3 bars -> 30 bars.
- Slow SMA: 1008 bars (21 days) -> 630 bars (13 days).
- Trail tightening ADDED: 4.24 -> 2.0 x ATR after 15 x ATR of favourable excursion.
- Pyramid add sizing: current-bar ATR -> ATR captured AT ENTRY.
- Entry logic, exit anchor, ATR period, add count, add spacing, short risk multiplier and cost model are UNCHANGED.

4. Complete Pine Script — Version 2 — Config E

Source: git 927f485:strategies/gold_trend_E.pine. Reproduced in full with no elisions. Inputs carry the exact defaults that reproduce the configuration documented above. Closed-bar only.

```
// =====
// GOLD TREND - CONFIG E (drawdown-budgeted champion)
// =====
// Identical engine to the champion. Four settings differ from the July
// build, each of which beat a leverage-matched control:
// 1. adds sized on the ATR AT ENTRY, not the current ATR
// 2. cooldown 30 bars (was 3, then 24)
// 3. slow SMA 13 days (was 21)
// 4. trail tightens 4.24 -> 2.0 ATR after a 15 ATR favourable run
//
// RISK IS SOLVED, NOT CHOSEN. The drawdown budget is the input; the risk
// percentage is whatever makes the historical max drawdown land on it.
//
// XAUUSD 30m, Dec 2019 - Jul 2026, 0.20 pt slippage/side, 711 trades
// budget risk PF net maxDD MAR MC med P(DD>30%)
// 25% 0.705% 1.945 +1,875% 24.98% 2.26 25.3% 21.5%
// 20% 0.553% 1.906 +998% 20.00% 2.17 20.4% 5.0%
//
// Splits: TRAIN PF 1.326 | VALIDATION 1.623 | TEST (untouched) 2.306
// US30, settings unchanged: PF 1.469
// Worst year 2021 at PF 0.94-1.00 (flat). Win rate ~21.7%.
//
// KNOW BEFORE TRADING IT
// - Longest losing run in the sample: 25 consecutive positions.
// - The top 10 trades are ~95% of net profit.
// - Break-even slippage is ~1.6 points. Set the value below to YOUR spread.
// - Whole-ounce sizing needs ~$25,000 at gold near $4,000 to behave as tested.
// =====
//@version=6
strategy("Gold Trend E - DD-budgeted",
  overlay = true,
  initial_capital = 10000,
  currency = currency.USD,
  default_qty_type = strategy.cash,
  default_qty_value = 1000,
  commission_type = strategy.commission.cash_per_contract,
  commission_value = 0.07,
  // SLIPPAGE IS IN TICKS, NOT POINTS. XAUUSD/OANDA quotes three decimals,
  // so mintick is 0.001 and 200 ticks = 0.20 points - a realistic retail
  // gold fill. The earlier value of 5 modelled 0.005 points, roughly 40x
  // too little, and it flattered every backtest run with it.
  // COST STRESS (full period, Balanced, measured 2026-08-02):
  // 0.05 pt ( 50) -> PF 1.751 +2,533% drawdown 27.7%
  // 0.20 pt (200) -> PF 1.641 +1,620% drawdown 33.4% <- shipped
  // 0.50 pt (500) -> PF 1.452 +715% drawdown 52.2%
  // 1.00 pt (1000)-> PF 1.141 +109% drawdown 76.1%
  // 1.50 pt (1500)-> PF 0.888 -54% BREAK-EVEN IS HERE
  // Set this to YOUR broker's actual gold spread before trusting any
  // number this script reports.
  slippage = 200,
  process_orders_on_close = true,
  // BUG-012 guard. Pine's margin default of 100% silently REJECTS
  // risk-sized orders with no error and no log line. It has bitten
  // this project twice.
  margin_long = 5,
  margin_short = 5,
  calc_on_every_tick = false,
  // REQUIRED by the add-to-winner block. With Pine's default of 0,
  // strategy.entry while a position is open is silently ignored.
  pyramiding = 5,
  // The regime EMA plus its slope lookback reaches ~2,200 bars back.
  // Pine's default buffer is 300, which yields na without an error.
  max_bars_back = 5000)

// — @ Auto-scale —————
// Every lookback here is in BARS, so changing timeframe silently doubles
// or halves it. That caused two failed reproductions, so the script
// derives its own settings from the chart instead.
// 15m -> entry 48, SMA 768, EMA 2016, slope 192, trail 6.00
// 30m -> entry 24, SMA 384, EMA 1008, slope 96, trail 4.24
bool autoScale = input.bool(true, "Auto-scale settings to the chart timeframe", group="@ Auto-scale",
  tooltip="ON is strongly recommended. The inputs below are then IGNORED and derived from the timeframe, so moving
  between 15m and 30m cannot silently change the strategy. The dashboard shows what is actually in use. RUN THIS
  ON 30m: it beat 15m on profit factor, return AND drawdown in both halves.")

// — ● RISK PROFILE (the only dial most people need) —————
// Same engine, same entry, same exit. Only how HARD it presses.
// Full 6.7 years, XAUUSD 30m, gap-aware fills:
//
// profile adds risk net maxDD worst yr PF
// Conservative 3 ATR x2 1.0% +660% 17.7% -1.5% 1.626
// Balanced 1.5 ATR x4 1.0% +2,307% 29.8% -6.3% 1.728
// Aggressive 1.5 ATR x4 1.5% +8,627% 42.8% -12.1% 1.769
// Maximum 1.5 ATR x4 2.0% +24,712% 54.5% -22.5% 1.764
//
// All four are positive in all five walk-forward slices and all four lose
// only in 2021. Return roughly triples at each step and so does the pain:
// Maximum spent 2021 down 22.5% with a 45.8% intra-year drawdown. Choose
// by the drawdown you can hold through, not by the return.
string riskProf = input.string("25% DD budget", "Risk setting", options=["25% DD budget", "20% DD budget", "Custom"],
  group="● Risk Profile", tooltip="Risk per trade is SOLVED so the historical maximum drawdown lands on the
  stated budget. It is not a preference dial - the drawdown target is the input and the risk percentage is the
  output.\n\n25% BUDGET: risk 0.705%. PF 1.945, net +1,875%, max DD 24.98%, MAR 2.26. Monte Carlo median DD
  25.3%, 21.5% chance of exceeding 30%.\n\n20% BUDGET: risk 0.553%. PF 1.906, net +998%, max DD 20.00%, MAR
  2.17. Monte Carlo median DD 20.4%, only 5.0% chance of exceeding 30% - the most trustworthy drawdown figure
  this project has produced.\n\nA backtest drawdown is NOT an expectation. Reordering the same trades gives a
  median about 1 point worse than the historical path. Size for the Monte Carlo median, not the backtest.")
// — ● Audit findings (2026-07-31 expert pass) —————
```



```
// 35 entry x exit combinations were measured after this build shipped.
// NOTHING beat it on the criteria that predict live behaviour (zero
// losing walk-forward slices, one losing year, best return-per-drawdown
// among the zero-loss options). Two real findings came out of it:
int coolDown = input.int(30, "Cooldown after an exit (bars)", minval=0, maxval=100, group="🔴 Audit Findings",
    tooltip="UPDATED 2026-08-03 from 3 to 24 bars (12 hours on 30m). Re-swept 6-48 under entry-ATR add sizing:
    return/drawdown runs 64/60/72/68/74/78/84/79/58/56/22 for cooldown 6/9/12/16/20/24/28/32/36/40/48, a broad
    plateau from 12 to 32 with the baseline at 62. 24 is the middle of that plateau, not its peak (28 scores
    highest). Confirmed on US30: ret/DD 6.73 vs 4.33.\n\nORIGINAL NOTE: after an exit the shipped build could re-
    enter on the very next bar. Swept 0/3/10/24/48 bars: 3 was the free improvement - full period +2,432% to
    +2,533% and drawdown 28.55% to 27.72% with no cost anywhere else. 24 bars gives the best out-of-sample profit
    factor (2.098) but less return; 48 costs a third of the return.")
bool longSlope = input.bool(false, "Longs ALSO require a rising EMA", group="🔴 Audit Findings",
    tooltip="AUDIT FINDING, defaulted OFF because it is a genuine trade-off, not a free win. Shorts must have the EMA
    falling but longs were gated on nothing - an asymmetry that was never tested until now.\n\nWith longs also
    requiring a RISING EMA: profit factor 1.746 to 1.778, out-of-sample 2.065 to 2.179, and the losing year
    DISAPPEARS - 0 of 7 years negative instead of 1 of 7.\n\nThe cost: one of the five walk-forward slices turns
    negative (0/5 becomes 1/5) and drawdown rises 28.55% to 33.78%. It trades a bad year for a bad quarter. Turn
    it on if a losing calendar year is what you cannot tolerate; leave it off if drawdown is.")

// — 🟢 Entry —————
string entryMode= input.string("Donchian breakout", "Entry trigger", options=["Donchian breakout", "HMA cross"],
    group="🟢 Entry",
    tooltip="Both measured through the identical exit, filters and adds.\n\nDONCHIAN: PF 1.626, +660.2%, drawdown
    17.68%, return/drawdown 37.3.\n\nHMA CROSS: PF 1.743, +540.4%, drawdown 20.9%, return/drawdown 25.9.\n\nHMA has
    the higher profit factor in BOTH halves (train 1.462 vs 1.370, out of sample 2.008 vs 1.879) - it is more
    accurate per trade. Donchian captures more and does it at less drawdown. Neither is strictly better.\n\nAlso
    measured and NOT offered: SMA and EMA crosses. The SMA 200 cross, the usual default everywhere, is the worst
    setting tested - PF 0.920 and -17.0% with no filters.")
int entryLen = input.int(50, "Breakout lookback (bars)", minval=5, group="🟢 Entry",
    tooltip="Long when close exceeds the highest high of the prior N bars. Validated at 50 and 100. A timing trigger,
    not an edge.")
int hmaLen = input.int(200, "HMA length (HMA mode only)", minval=10, group="🟢 Entry",
    tooltip="Swept 50/100/200/400/800; 200 best by the weaker of the two halves. NOT auto-scaled - it was measured at
    200 on 30m specifically.")
bool longOnly = input.bool(false, "Long only (leave OFF)", group="🟢 Entry",
    tooltip="THIS IS A REGIME BET, NOT A SETTING. Long-only wins every rising year and loses in falling ones. In the
    Feb-Jul 2026 fall, long-only returned -5.72% live while long+short returned +7.57%.")

// — 🟢 Risk and filters —————
float riskPct = input.float(1.0, "Risk per trade (%)", step=0.25, group="🟢 Risk & Filters",
    tooltip="The direct return lever, and it is not free. Measured: 1% -> +85.4% train at 13.6% drawdown; 2% ->
    +208.8% at 28.6%; 3% -> +367.4% at 41.8%. Return scales roughly linearly and so does drawdown. Choose by the
    drawdown you can sit through.")
float maxLever = input.float(20, "Max leverage (notional cap)", minval=1, group="🟢 Risk & Filters",
    tooltip="BUG-012 guard. Without a notional cap, risk-based sizing can request more than buying power and
    TradingView skips the order with no error.")
float trailMult = input.float(6.0, "Trailing stop (x ATR, 15m basis)", minval=1.0, step=0.5, group="🟢 Risk & Filters",
    tooltip="THE key parameter, and the one whose old description was wrong. Wider raises PROFIT FACTOR but LOWERS
    RETURN, because risk-based sizing shrinks the position as the stop widens: trail 20 ATR gives PF 1.659 but
    only +23.5%, against trail 6's 1.243 and +85.4%. 6 on a 15m basis (4.24 on 30m) is return-optimal at tolerable
    drawdown.")
int atrLen = input.int(14, "ATR length", minval=2, group="🟢 Risk & Filters")
bool useSma = input.bool(true, "Fast SMA must agree", group="🟢 Risk & Filters",
    tooltip="750 bars on 15m, about 8 days. Selected on the training half only.")
int smaLen = input.int(750, "Fast SMA length (bars)", minval=50, group="🟢 Risk & Filters")
bool useSma2 = input.bool(true, "Slow SMA must agree TOO (the confluence layer)", group="🟢 Risk & Filters",
    tooltip="Requiring BOTH averages lifts full-period PF 1.379 -> 1.421 and cuts drawdown 12.30% -> 9.63%. Return
    barely moved; the drawdown fell by a fifth. It is a risk filter, not a return generator.\n\nIt is the
    AGREEMENT that works, not the slower average: SMA2000 instead of SMA750 is worse (train 1.181 vs 1.289), and
    requiring only ONE of the two is much worse (1.149). The two horizons disagreeing marks the choppy middle of a
    range, and that is what gets skipped.")
int sma2Len = input.int(2000, "Slow SMA length (bars)", minval=100, group="🟢 Risk & Filters")
bool useRegime = input.bool(true, "EMA regime gate", group="🟢 Risk & Filters",
    tooltip="Only long above the EMA, only short below it.")
int regimeLen = input.int(2000, "Regime EMA length (bars)", minval=20, group="🟢 Risk & Filters")
bool useSlope = input.bool(true, "Shorts only when the EMA is FALLING", group="🟢 Risk & Filters",
    tooltip="This is what makes the system profitable in a falling market without giving up the trend returns. Price
    sits below a FLAT EMA constantly during uptrend pullbacks; the EMA actually declining is far rarer and is what
    a bear regime looks like. Gating shorts on slope keeps them nearly silent in bull markets, where counter-
    trend shorts are what bleed away the edge.")
int slopeLen = input.int(200, "Slope lookback (bars)", minval=20, group="🟢 Risk & Filters")
float shortRisk = input.float(0.75, "Short risk multiplier", minval=0.1, maxval=3.0, step=0.05, group="🟢 Risk &
    Filters",
    tooltip="Counter-trend trades sized smaller. 0.75 gave the best full-period result; raise toward 1.5 for more
    downside protection at the cost of total return.")

// — 🟢 Add to winners —————
bool usePyr = input.bool(true, "Add to a winning position", group="🟢 Add to Winners",
    tooltip="THE BIGGEST RETURN LEVER MEASURED HERE, and it multiplies BOTH sides. Walk-forward over five consecutive
    slices, net/drawdown, without adds vs with 2 adds every 3 ATR:\n\n +38.4/9.2 -> +92.5/16.3\n +19.6/6.7
    -> +23.9/13.5\n +8.8/10.4 -> +11.9/16.6\n +15.1/10.2 -> +33.3/16.8\n +28.4/7.8 -> +88.1/13.5\n\nEvery
    slice improves, which is why it ships on. But through RANDOM entries it lifts the median from 6.8% to 25.3%
    while taking the spread from 6.8 to 29.5 and the worst seed from -3.5% to -30.2%. It is not an edge. It is
    betting more on trades that are already working.")
float pyrATR = input.float(3.0, "Add every (x ATR of favourable move)", minval=0.5, step=0.5, group="🟢 Add to
    Winners",
    tooltip="3.0 with 2 adds is the moderate setting. 2.0 with 4 adds returns far more (+163% and +168% on the first
    and last slices) at drawdowns up to 26.4%.")
int pyrMax = input.int(2, "Maximum adds", minval=1, maxval=4, group="🟢 Add to Winners")
float pyrRisk = input.float(1.0, "Each add sized at (x original risk)", minval=0.1, maxval=1.0, step=0.1, group="🟢
    Add to Winners")

// — 🟢 Trailing algorithm —————
string trailMode= input.string("Chandelier (ATR)", "Trailing logic", options=["Chandelier (ATR)", "Donchian low"],
    group="🟢 Trailing Algorithm",
    tooltip="Six algorithms were built and measured: chandelier, Donchian low, EMA, a multiple that tightens with
    profit, a give-back-a-fraction trail, and none. Two are worth offering.\n\nCHANDELIER (default): PF 1.635,
    +679.9%, drawdown 17.01%, positive in ALL FIVE walk-forward slices.\n\nDONCHIAN LOW: PF 1.624, +935.9%, drawdown
    23.45%, NEGATIVE in one of five (-6.0%).\n\nDonchian returns 38% more, but return per drawdown is a dead heat
    (39.9 vs 40.0) - the same efficiency at larger size, with one losing period to sit through.\n\nRejected: EMA
    trail (best PF 1.383 at 33.7% drawdown), give-back trail (1.061), stepping trail (1.597, behind both.")
float tightAfter= input.float(15.0, "Tighten the trail after this much favourable move (x ATR at entry)", minval=0,
    step=1.0, group="🟢 Trailing Algorithm",
    tooltip="0 disables. PRIORITY 4 FINDING (2026-08-03). Once a trade has run this far in its favour, the leash
    tightens to the value below. This is NOT a tight stop and NOT a profit target: it only ever engages on trades
```

```

already deep in profit, and the trade still exits on a trail.\n\nAt 20 ATR -> 2.0: profit factor 1.714 ->
1.834, net +3,855% -> +5,922%, drawdown 32.65% -> 33.80%. Against a leverage-matched baseline at the same
drawdown it returns +5,922% vs about +4,500% with a higher profit factor, so it is not the risk dial in
disguise. Improves the profit factor in 5 of 8 calendar years. Confirmed on US30: PF 1.469 vs 1.351 at
identical drawdown.\n\nSwept 10-30 ATR x 1.5-3.0: 21 of 28 combinations beat the baseline and everything from
18 ATR up is a plateau. 20 is inside it, not its peak.")
float tightTo = input.float(2.0, "...tighten it to (x ATR)", minval=0.5, step=0.25, group="Ⓢ Trailing Algorithm")
int donchN = input.int(45, "Donchian trail lookback (bars)", minval=5, group="Ⓢ Trailing Algorithm",
    tooltip="The stop rides the lowest low of the last N bars for longs. Do NOT read the training peak at 40 as
    meaningful: out of sample the whole 30-50 range performs and the ordering reverses, which is what a fitted
    spike looks like once you test it. 45 is the middle of the out-of-sample plateau, not the training maximum.")

// — Ⓢ TP1 / TP2 (staged profit taking, OFF by default)
bool tp1On = input.bool(false, "TP1 - bank part of the position", group="Ⓢ TP1 / TP2",
    tooltip="MEASURED 20 WAYS AND IT COSTS RETURN EVERY TIME. Best variant: +234.1% against +253.1% with no target.
    Adding TP2 as well: +216.7%. Out of sample the gap widens - PF 1.568 with TP1 against 1.930 without. Even
    restricting targets to the counter-trend side fails (+242.6%), the only place in this project where that
    asymmetry rule has not held.\n\nWhat it DOES buy is a smoother curve: full-period drawdown 17.01% -> 16.04%,
    out-of-sample drawdown 12.79% -> 8.38%. A real trade, just not a free one.")
float tp1Atr = input.float(4.0, "TP1 distance (x ATR)", minval=0.5, step=0.5, group="Ⓢ TP1 / TP2")
float tp1Pct = input.float(25, "TP1 - % of position closed", minval=5, maxval=90, step=5, group="Ⓢ TP1 / TP2")
bool tp2On = input.bool(false, "TP2 - bank another part", group="Ⓢ TP1 / TP2")
float tp2Atr = input.float(12.0, "TP2 distance (x ATR)", minval=1.0, step=0.5, group="Ⓢ TP1 / TP2")
float tp2Pct = input.float(25, "TP2 - % of position closed", minval=5, maxval=90, step=5, group="Ⓢ TP1 / TP2")
bool tpShorts = input.bool(false, "Apply TP1/TP2 to shorts only", group="Ⓢ TP1 / TP2")
// — Ⓢ Pure profit-taking mode (different architecture)
string tpMode = input.string("Off", "Fixed-target mode", options=["Off", "Percent of price", "Fixed points"],
    group="Ⓢ Profit-Taking Mode",
    tooltip="TURNS THIS INTO A PROFIT-TAKING STRATEGY instead of a trend-following one. The trail and the adds are
    bypassed: entry, static stop, fixed target, done.\n\nMeasured with a 4 ATR stop and a 4% target: train PF
    1.402 / +98.5%, out of sample PF 1.420 / +23.2%, full period PF 1.475 / +185.3% at 23.2% drawdown. Train and
    out of sample agree to 0.018 of profit factor - the most stable result in this project.\n\nIt beats the
    trailing build on profit factor (1.475 vs 1.444 without adds) and loses badly on drawdown and return. Use it
    if you want frequent closed wins; leave it Off if you want the return.\n\nUSE PERCENT, NOT POINTS: gold ran
    1450 -> 4100 over the sample, so a fixed 40-point target was 2.7% of price at the start and 1.0% at the end.
    Percent scored PF 1.475 against points' 1.160.")
float tpPct = input.float(4.0, "Target - percent of price", minval=0.25, step=0.25, group="Ⓢ Profit-Taking Mode",
    tooltip="Swept 1.5/2/3/4%. Bigger is monotonically better on profit factor; 4% is where drawdown stops
    improving.")
float tpPoints = input.float(160, "Target - fixed points", minval=1, step=5, group="Ⓢ Profit-Taking Mode")
float tpStopAtr = input.float(4.0, "Static stop for this mode (x ATR)", minval=0.5, step=0.5, group="Ⓢ Profit-Taking
    Mode",
    tooltip="Swept 2/3/4/6/8 ATR. 4 was best; 2 was catastrophic (43% drawdown) because the target then needs an
    unreachable hit rate.")

// — Ⓢ Display
bool showStop = input.bool(true, "Plot the stop", group="Ⓢ Display")
bool showBand = input.bool(true, "Plot the entry trigger", group="Ⓢ Display")
bool showDash = input.bool(true, "Show dashboard", group="Ⓢ Display")

// =====
// EFFECTIVE SETTINGS
// Targets: breakout ~12h, fast SMA ~8 days, EMA ~21 days, slope ~2 days.
// The trail scales with the SQUARE ROOT of the timeframe because ATR
// grows roughly as sqrt(time); that rule was validated by sweeping the
// 30m trail directly (4.24 was best on train net, +113.5%).
// =====
// Profile presets override the manual risk and add settings.
float riskE = riskProf == "20% DD budget" ? 0.553 : riskProf == "Custom" ? riskPct : 0.705
float pyrAtrE = riskProf == "Custom" ? pyrAtr : 1.5
int pyrMaxE = riskProf == "Custom" ? pyrMax : 4

float tfMin = timeframe.in_seconds() / 60.0
int entryLenE = autoScale ? math.max(5, int(12.0 * 60.0 / tfMin)) : entryLen
int smaLenE = autoScale ? math.max(50, int(8.0 * 24.0 * 60.0 / tfMin)) : smaLen
int regimeLenE = autoScale ? math.max(20, int(21.0 * 24.0 * 60.0 / tfMin)) : regimeLen
int sma2LenE = autoScale ? math.max(100, int(13.0 * 24.0 * 60.0 / tfMin)) : sma2Len
int slopeLenE = autoScale ? math.max(20, int(2.0 * 24.0 * 60.0 / tfMin)) : slopeLen
float trailMultE = autoScale ? 6.0 * math.sqrt(15.0 / tfMin) : trailMult

// =====
// SIGNALS - every ta.* call unconditional at top level (BUG-004)
// =====
float atr = ta.atr(atrLen)
float donHi = ta.highest(high, entryLenE)[1]
float donLo = ta.lowest(low, entryLenE)[1]
float donLoTrail = ta.lowest(low, donchN)[1]
float donHiTrail = ta.highest(high, donchN)[1]
float regimeMa = ta.ema(close, regimeLenE)
float trendSma = ta.sma(close, smaLenE)
float slowSma = ta.sma(close, sma2LenE)
float hmaVal = ta.hma(close, hmaLen)
bool hmaUp = ta.crossover(close, hmaVal)
bool hmaDn = ta.crossunder(close, hmaVal)

bool emaFalling = regimeMa < regimeMa[slopeLenE]
bool emaRising = regimeMa > regimeMa[slopeLenE]
bool regimeOkL = (not useRegime or close > regimeMa) and (not longSlope or emaRising)
bool regimeOkS = (not useRegime or close < regimeMa) and (not useSlope or emaFalling)
bool smaOkL = not useSma or (not na(trendSma) and close > trendSma)
bool smaOkS = not useSma or (not na(trendSma) and close < trendSma)
bool sma2OkL = not useSma2 or (not na(slowSma) and close > slowSma)
bool sma2OkS = not useSma2 or (not na(slowSma) and close < slowSma)

bool trigUp = entryMode == "HMA cross" ? hmaUp : (not na(donHi) and close > donHi)
bool trigDn = entryMode == "HMA cross" ? hmaDn : (not na(donLo) and close < donLo)

bool longSig = trigUp and regimeOkL and smaOkL and sma2OkL
bool shortSig = not longOnly and trigDn and regimeOkS and smaOkS and sma2OkS

// =====
// STATE
// =====
bool tpOn = tpMode != "Off"

var int barsSinceExit = 0

```

```

barsSinceExit := strategy.position_size == 0 ? barsSinceExit + 1 : 0
bool coolOk = barsSinceExit > coolDown

var float trailStop = na
var float tpStop    = na
var float tpTarget  = na
var float tp1Px     = na
var float tp2Px     = na
var bool  tp1Done   = false
var bool  tp2Done   = false
var float lastAdd    = na
var int   nAdds      = 0
// ATR AT ENTRY, for add sizing. PRIORITY 2 FINDING (2026-08-03): sizing each
// add off the CURRENT ATR shrinks it exactly when volatility is expanding,
// which on this system is precisely when the trade is working. Sizing off the
// entry ATR - equivalently, every add equals the ORIGINAL position size -
// scored PF 1.647 vs 1.623, net +1,996% vs +1,617% and drawdown 32.2% vs
// 36.5%. Against a leverage-matched baseline at the same drawdown it returned
// +1,996% vs +1,094% with a higher PF, so it is not the risk dial in disguise.
var float atrEntry = na
var float bestPx   = na // best excursion THROUGH THE PREVIOUS BAR (see below)

// =====
// TRAILING STOP
// Ratchets in the trade's favour only, anchored to the PREVIOUS bar's
// extreme so the current bar can never move its own stop and then hit it.
// =====
// Excursion is measured off bestPx, which holds the extreme through the
// PREVIOUS bar only. Using this bar's own high would let the bar tighten its
// own stop and then hit it -- the intrabar-lookahead defect of BUG-019.
float excAtr = strategy.position_size == 0 or na(atrEntry) or na(bestPx) or atrEntry <= 0 ? 0.0 :
    (strategy.position_size > 0 ? bestPx - strategy.position_avg_price
     : strategy.position_avg_price - bestPx) / atrEntry
float effTrail = tightAfter > 0 and excAtr >= tightAfter ? tightTo : trailMultE

if strategy.position_size > 0
    float cand = trailMode == "Donchian low" ? donLoTrail : high[1] - atr * effTrail
    trailStop := na(trailStop) ? cand : math.max(trailStop, cand)
else if strategy.position_size < 0
    float cand = trailMode == "Donchian low" ? donHiTrail : low[1] + atr * effTrail
    trailStop := na(trailStop) ? cand : math.min(trailStop, cand)
else
    trailStop := na
    lastAdd   := na
    atrEntry  := na
    bestPx    := na
    nAdds     := 0

// =====
// ADD TO A WINNING POSITION
// PORT DELTA: with process_orders_on_close the add fills at THIS bar's
// close, so the enlarged position only meets the trailing stop from the
// next bar. The research engine adds at the trigger price and exposes the
// enlarged size on the same bar, so Python is the stricter of the two.
// =====
if usePyr and not tpOn and strategy.position_size != 0 and nAdds < pyrMaxE and not na(lastAdd)
    float step = atr * pyrAtrE
    bool  isLong = strategy.position_size > 0
    float nxt   = isLong ? lastAdd + step : lastAdd - step
    if isLong ? high >= nxt : low <= nxt
        float aRisk = (isLong ? riskE : riskE * shortRisk) * pyrRisk
        float aAtr  = na(atrEntry) ? atr : atrEntry // entry-ATR sizing
        float aQty  = math.floor(math.min(strategy.equity * aRisk / 100.0 / (aAtr * trailMultE),
            strategy.equity * maxLev / close - math.abs(strategy.position_size)))
        if aQty >= 1
            strategy.entry(isLong ? "L" : "S", isLong ? strategy.long : strategy.short, qty=aQty)
            nAdds := nAdds + 1
            lastAdd := nxt

// =====
// ENTRIES
// =====
bool wantShort = shortSig and not longSig
float effRisk  = wantShort ? riskE * shortRisk : riskE
float stopDist = atr * trailMultE
float qtyRaw   = stopDist > 0 ? strategy.equity * effRisk / 100.0 / stopDist : 0
float qtyCap   = strategy.equity * maxLev / close
float qty      = math.floor(math.min(qtyRaw, qtyCap))

if longSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("L", strategy.long, qty=qty)
    trailStop := close - stopDist
    tpStop    := close - atr * tpStopAtr
    tpTarget   := tpMode == "Fixed points" ? close + tpPoints : close * (1.0 + tpPct / 100.0)
    tp1Px     := close + atr * tp1Atr
    tp2Px     := close + atr * tp2Atr
    tp1Done   := false
    tp2Done   := false
    lastAdd   := close
    atrEntry  := atr
    bestPx    := close
    nAdds     := 0

if shortSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("S", strategy.short, qty=qty)
    trailStop := close + stopDist
    tpStop    := close + atr * tpStopAtr
    tpTarget   := tpMode == "Fixed points" ? close - tpPoints : close * (1.0 - tpPct / 100.0)
    tp1Px     := close - atr * tp1Atr
    tp2Px     := close - atr * tp2Atr
    tp1Done   := false
    tp2Done   := false
    lastAdd   := close
    atrEntry  := atr
    bestPx    := close
    nAdds     := 0

```

```

nAdds      := 0

// =====
// STAGED PROFIT TAKING (off by default)
// PORT DELTA: these fill at the bar CLOSE via strategy.close, not at the
// target price. Two live limit orders plus a trailing stop on one entry
// is where order conflicts come from. The research engine fills at the
// target, so expect Pine to read slightly worse here.
// =====
bool tpSideOk = not tpShorts or strategy.position_size < 0
bool tp1Hit = tp1On and not tpOn and not tp1Done and tpSideOk and strategy.position_size != 0 and
  (strategy.position_size > 0 ? high >= tp1Px : low <= tp1Px)
bool tp2Hit = tp2On and not tpOn and not tp2Done and tpSideOk and strategy.position_size != 0 and
  (strategy.position_size > 0 ? high >= tp2Px : low <= tp2Px)

if tp1Hit
  strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp1Pct, comment="TP1")
  tp1Done := true
if tp2Hit
  strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp2Pct, comment="TP2")
  tp2Done := true

// =====
// EXITS
// Fixed-target mode replaces the ratchet entirely: static stop, fixed
// limit, no trail and no adds. Anything else would be a third
// architecture that was never measured.
// =====
if tpOn
  if strategy.position_size > 0 and not na(tpStop)
    strategy.exit("L-exit", "L", stop=tpStop, limit=tpTarget)
  if strategy.position_size < 0 and not na(tpStop)
    strategy.exit("S-exit", "S", stop=tpStop, limit=tpTarget)
else
  if strategy.position_size > 0 and not na(trailStop)
    strategy.exit("L-exit", "L", stop=trailStop)
  if strategy.position_size < 0 and not na(trailStop)
    strategy.exit("S-exit", "S", stop=trailStop)

// Roll bestPx forward only now that this bar's stop and exits are resolved,
// so the value the next bar reads is the extreme through THIS bar and no
// bar can ever act on its own high.
if strategy.position_size != 0
  bestPx := na(bestPx) ? close : (strategy.position_size > 0 ? math.max(bestPx, high) : math.min(bestPx, low))

// =====
// VISUALS
// =====
plot(showBand and entryMode == "Donchian breakout" and not na(donHi) ? donHi : na, "Breakout high",
  color.new(color.orange, 40), 1, plot.style_stepline)
plot(showBand and entryMode == "Donchian breakout" and not na(donLo) ? donLo : na, "Breakout low",
  color.new(color.orange, 70), 1, plot.style_stepline)
plot(showBand and entryMode == "HMA cross" ? hmaVal : na, "HMA", color.new(color.fuchsia, 0), 2)
plot(showStop and strategy.position_size != 0 ? (tpOn ? tpStop : trailStop) : na, "Stop", color.new(#D50000, 0), 2,
  plot.style_stepline)
plot(tpOn and strategy.position_size != 0 ? tpTarget : na, "Fixed target", color.new(#00E676, 0), 2,
  plot.style_stepline)
plot(tp1On and not tpOn and strategy.position_size != 0 and not tp1Done ? tp1Px : na, "TP1", color.new(#00E676, 20),
  1, plot.style_circles)
plot(tp2On and not tpOn and strategy.position_size != 0 and not tp2Done ? tp2Px : na, "TP2", color.new(#00E676, 45),
  1, plot.style_circles)
plot(useRegime ? regimeMa : na, "Regime EMA", color.new(color.fuchsia, 55), 1)
plot(useSma ? trendsma : na, "Fast SMA", color.new(color.aqua, 25), 2)
plot(useSma2 ? slowSma : na, "Slow SMA", color.new(#7E57C2, 15), 2)

plotshape(longSig and strategy.position_size == 0, "Long entry", shape.triangleup, location.belowbar,
  color.new(#00C853, 0), size=size.small, text="LONG")
plotshape(shortSig and strategy.position_size == 0, "Short entry", shape.triangledown, location.abovebar,
  color.new(#D50000, 0), size=size.small, text="SHORT")

bool exitedLong = strategy.position_size[1] > 0 and strategy.position_size == 0
bool exitedShort = strategy.position_size[1] < 0 and strategy.position_size == 0
plotshape(exitedLong or exitedShort, "Exit", shape.xcross, location.absolute, color.new(color.gray, 0),
  size=size.tiny)

// =====
// ALERTS
// =====
alertcondition(longSig, "Gold Trend - LONG entry", "LONG {{ticker}} {{interval}} at {{close}}")
alertcondition(shortSig, "Gold Trend - SHORT entry", "SHORT {{ticker}} {{interval}} at {{close}}")
alertcondition(exitedLong or exitedShort, "Gold Trend - position closed", "EXIT {{ticker}} {{interval}} at {{close}}")
alertcondition(tp1Hit, "Gold Trend - TP1 hit", "TP1 {{ticker}} {{interval}} at {{close}}")
alertcondition(tp2Hit, "Gold Trend - TP2 hit", "TP2 {{ticker}} {{interval}} at {{close}}")

// =====
// DASHBOARD
// =====
var table dash = table.new(position.top_right, 2, 10, bgcolor=color.new(color.black, 20), border_width=1,
  border_color=color.new(color.gray, 60))
if showDash and barstate.islast
  int nT = strategy.closedtrades
  int wT = 0
  float gw = 0.0
  float gl = 0.0
  if nT > 0
    for i = 0 to nT - 1
      float p = strategy.closedtrades.profit(i)
      wT += p > 0 ? 1 : 0
      gw += p > 0 ? p : 0.0
      gl += p < 0 ? -p : 0.0
  float wr = nT > 0 ? wT * 100.0 / nT : na
  float pf = gl > 0 ? gw / gl : na
  float netPct = (strategy.equity - strategy.initial_capital) / strategy.initial_capital * 100
  int nUp = (regimeOkL ? 1 : 0) + (smaOkL ? 1 : 0) + (sma2OkL ? 1 : 0)
  int nDn = (regimeOkS ? 1 : 0) + (smaOkS ? 1 : 0) + (sma2OkS ? 1 : 0)

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table.cell(dash, 0, 0, "GOLD TREND", text_color=color.orange, text_size=size.small)
table.cell(dash, 1, 0, riskProf + " " + str.tostring(riskE, "#.#") + "%", text_color=color.orange,
text_size=size.small)
table.cell(dash, 0, 1, "Trades", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 1, str.tostring(nT), text_color=nT < 250 ? color.yellow : color.white, text_size=size.tiny)
table.cell(dash, 0, 2, "Win rate", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 2, na(wr) ? "-" : str.tostring(wr, "#.#") + "%", text_color=color.white, text_size=size.tiny)
table.cell(dash, 0, 3, "Profit factor", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 3, na(pf) ? "-" : str.tostring(pf, "#.##"), text_color=na(pf) ? color.gray : pf >= 1.3 ?
color.lime : pf >= 1.0 ? color.yellow : color.red, text_size=size.tiny)
table.cell(dash, 0, 4, "Net", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 4, str.tostring(netPct, "#.#") + "%", text_color=netPct > 0 ? color.lime : color.red,
text_size=size.tiny)
table.cell(dash, 0, 5, "Position", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 5, strategy.position_size > 0 ? "LONG +" + str.tostring(nAdds) : strategy.position_size < 0 ?
"SHORT +" + str.tostring(nAdds) : "flat", text_color=strategy.position_size != 0 ? color.lime : color.gray,
text_size=size.tiny)
table.cell(dash, 0, 6, "Stop", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 6, na(trailStop) and na(tpStop) ? "-" : str.tostring(tpOn ? tpStop : trailStop,
format.mintick), text_color=color.red, text_size=size.tiny)
table.cell(dash, 0, 7, "Confluence", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 7, nUp == 3 ? "LONG 3/3" : nDn == 3 ? "SHORT 3/3" : "none (" + str.tostring(math.max(nUp,
nDn)) + "/3)", text_color=nUp == 3 ? color.lime : nDn == 3 ? color.red : color.gray, text_size=size.tiny)
table.cell(dash, 0, 8, "Exit engine", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 8, tpOn ? "fixed target" : trailMode == "Donchian low" ? "donchian " + str.tostring(donchN) :
"chandelier " + str.tostring(trailMultE, "#.##"), text_color=color.aqua, text_size=size.tiny)
table.cell(dash, 0, 9, autoScale ? "auto " + timeframe.period : "MANUAL", text_color=autoScale ? color.aqua :
color.orange, text_size=size.tiny)
table.cell(dash, 1, 9, autoScale ? "sma " + str.tostring(smaLenE) + "/" + str.tostring(sma2LenE) : (nT < 250 ?
"n<250: unstable" : ""), text_color=color.silver, text_size=size.tiny)

```

— end of Pine source (451 lines) —

Version 3 — Config G1

1. Identity and provenance

Field	Value
Label in research	Config G1
Pine file	strategies/gold_trend_G1.pine (revision at commit 927f485)
Commit	927f485
Change vs previous	Adds a regime risk multiplier applied to the OPENING position only.
Was it shipped/live?	Yes.

2. Complete rule specification

2.1 Entry

Field	Value
Bar close only	Yes. Every condition is evaluated on the CLOSED bar. No intrabar evaluation, no varip, no request.security with loc
Donchian breakout lookback	24 bars = 12.0 hours on 30m. Pine: entryLenE = max(5, int(12.0*60/tfMin)). Applied with a [1] offset: donHi = ta.hig
Regime EMA	EMA(close, 1008 bars) = 21.0 days on 30m. Pine: regimeLenE = max(100, int(21.0*24*60/tfMin)). Bar-based, not c
Fast SMA	SMA(close, 384 bars) = 8.0 days on 30m. Pine: smaLenE = max(50, int(8.0*24*60/tfMin)).
Slow SMA	SMA(close, 630 bars) = 13.0 days
Slope lookback	96 bars = 2.0 days on 30m. Pine: slopeLenE = max(20, int(2.0*24*60/tfMin)).
LONG conditions (all required)	1) close > ta.highest(high,24)[1] 2) close > EMA(close,1008) 3) close > SMA(close,384) 4) close > SMA(close,S
SHORT conditions (all required)	1) close < ta.lowest(low,24)[1] 2) close < EMA(close,1008) 3) close < SMA(close,384) 4) close < SMA(close,SL
Other entry filters	NONE. No session filter, no volume filter, no volatility filter, no candlestick or level confluence.
Cooldown after an exit	30 bars — a new entry requires barsSinceExit > cooldown.

2.2 Exit

Field	Value
Initial trail multiple	4.24 x ATR(14) initially, tightening to 2.0.
Number of exits	Exactly one: a ratcheting chandelier trailing stop.
ATR period	ATR(14), Wilder smoothing (RMA / ewm alpha=1/14).
Tightening rule	Identical to E: 4.24 -> 2.0 x ATR after 15 x ATR(entry) of excursion.
Trail anchor	PREVIOUS bar's extreme. Long: high[1] - atr*mult. Short: low[1] + atr*mult. A bar can never move its own stop and
Ratchet	Monotone in the trade's favour only. Long: trailStop := max(trailStop, cand). Short: min(trailStop, cand).
Profit target	NONE.
Time exit	NONE.
Breakeven stop	NONE.
Max holding period	Unbounded.

2.3 Pyramiding

Field	Value
Maximum adds	4
Add trigger spacing	every 1.5 x ATR of favourable movement
Add size formula	aQty = equity * (riskE * shortRiskIfShort * pyrRisk) / 100 / (ATR_AT_ENTRY * 4.24)
Regime multiplier on adds	NO. The multiplier scales the OPENING unit only. Adds are sized at base risk. This is the single difference from G1
Open risk treatment	Unbounded by design.

2.4 Position sizing and risk

Field	Value
Risk base	strategy.equity
Size formula	qty = floor(min(equity * effRisk / 100 / stopDist, equity * 20 / close)) where effRisk = (wantShort ? riskE * 0.75 : riskE
stopDist	ATR(14) * 4.24

Field	Value
Short risk multiplier	0.75
Regime multiplier — trend factor f1	trendStr = close - EMA(close,1008) / ATR(14). f1 = 1.4 if trendStr > 6.00 else 0.8
Regime multiplier — volatility factor f2	volExp = SMA(ATR(14),20) / SMA(ATR(14),200). f2 = 1.4 if volExp > 1.05 else 0.6
Regime multiplier — combination	regimeMult = clip(f1 * f2, 0.5, 2.0). Attainable values: 0.8*0.6=0.48 -> clipped to 0.50; 0.8*1.4=1.12; 1.4*0.6=0.84; 1
Risk solved to 25% DD budget	0.724% (Pine input default 0.723)
Risk solved to 20% DD budget	0.567%

2.5 Costs and execution assumptions used in research

Field	Value
Commission	USD 0.07 per contract (ounce) per side
Slippage	0.20 price points per side, applied against the trade
Notional cap	20x equity (maxLev). qty is capped at equity*20/close.
Overnight financing	NOT MODELLED. At ~38.7% time in market and ~2.6x time-weighted leverage, financing at 2-8% annual is an esti
Fill model	Entry at the signal bar's close. Stops are gap-aware (filled at the open if the bar gapped through). A bar touching bo

3. Full performance record

Gold XAUUSD 30m, 78,695 bars, 2019-12-01 to 2026-07-30 (6.66 years). Costs applied. Risk% is the SOLVED value that produced the stated drawdown.

Metric	25% DD budget	20% DD budget
Risk per trade (solved)	0.724%	0.567%
Trades (total)	711	711
Trades long	467	467
Trades short	244	244
Win rate	21.52%	21.52%
Average R	1.453	1.157
Profit factor	2.005	1.967
Net return	2,460.5%	1,250.6%
CAGR	62.75%	47.84%
Max drawdown (historical)	25.00%	20.00%
MAR (CAGR / MaxDD)	2.510	2.392
Avg hold (30m bars)	40.8	40.8
Avg hold (days)	0.85	0.85
Avg adds per trade	1.809	1.809
Avg winning trade (points)	33.02	33.02
Avg losing trade (points)	13.79	13.79
Monte Carlo median DD	24.86%	19.96%
Monte Carlo 95th pct DD	35.95%	29.31%
P(DD > 30%)	19.28%	4.12%
P(DD > 30%) std dev across seeds	0.301	0.114
Profit factor, pre-2025	1.513	1.528
PF TRAIN (first third)	1.555	1.596
PF VALID (middle third)	1.354	1.360
PF TEST (final third)	2.292	2.315
PF first half	1.370	1.404
PF second half	2.193	2.190
Top 5 trades, % of NET profit	47.6%	41.5%
Top 10 trades, % of NET profit	80.2%	70.1%
Top 5 trades, % of GROSS profit	23.9%	20.4%
Top 10 trades, % of GROSS profit	40.2%	34.5%

3.1 Year by year (25% drawdown budget)

Year	2019	2020	2021	2022	2023	2024	2025	2026
Trades	3	96	114	112	112	107	108	59
Profit factor	14.809	2.172	0.884	1.233	1.633	1.701	2.365	2.376

Down/flat years in this sample are 2021 (gold -4.3%), 2022 (-0.3%) and 2026 (-6.0%).

3.2 Anchored walk-forward, 6 folds (25% drawdown budget)

Fold	1	2	3	4	5	6
Profit factor	0.997	2.004	1.282	2.034	2.557	2.297

3.3 Reference: same configuration at a flat 1.0% risk

Risk	Trades	Profit factor	Net return	Max DD
1.000%	711	2.047	7,256.5%	33.31%

5. Differences from the previous version

- Regime risk multiplier ADDED, applied to the OPENING position only.
- Five new inputs: trendThr=6.0, expThr=1.05, upMult=1.4, dnMultT=0.8, dnMultV=0.6.
- Risk preset re-solved: 0.705 -> 0.723 (25% budget), 0.553 -> 0.567 (20%).
- Entry, exit, trail, tightening, add count, add spacing and add sizing are UNCHANGED from E.

4. Complete Pine Script — Version 3 — Config G1

Source: git 927f485:strategies/gold_trend_G1.pine. Reproduced in full with no elisions. Inputs carry the exact defaults that reproduce the configuration documented above. Closed-bar only.

```
// =====
// GOLD TREND - CONFIG G1 (config E + regime-scaled entry risk)
// =====
// Config E in every respect except one: the risk on a NEW ENTRY is scaled
// by how far price sits from its regime mean AND whether volatility is
// expanding. Press in strong expanding trends, shrink otherwise.
//
// XAUUSD 30m, Dec 2019 - Jul 2026, 0.20 pt slippage/side, 711 trades
// budget risk PF net maxDD MAR MC med P(DD>30%)
// 25% 0.723% 2.005 +2,457% 24.99% 2.51 24.8% 19.0%
// 20% 0.567% 1.967 +1,250% 20.00% 2.39 20.0% 5.0%
//
// Against config E it is better on EVERY metric on gold at both budgets,
// including a LOWER Monte Carlo median and tail - improving the drawdown
// distribution, not just the single historical path.
//
// WHY IT IS BELIEVED
// The same effect appears through three unrelated definitions (trend
// distance, ATR expansion ratio, ATR percentile) and is a plateau, not a
// spike: trend thresholds 3/4/5/6/7/8/10 ATR all beat config E.
//
// THE QUALIFIER, STATED PLAINLY
// On US30 with settings unchanged G1 scores PF 1.472 against E's 1.469 -
// NEUTRAL, not confirmed. Every previously adopted change showed a clear
// same-sign gain on the second instrument; this one does not. It also
// makes 2021 marginally worse (PF 0.89 against E's 0.94).
// About 85 configurations were searched to find a 3% profit-factor gain,
// which is close to what noise alone produces at that search count.
// TRADE E IF YOU WANT THE SETTLED VERSION; TRADE G1 IF YOU ACCEPT A
// THINNER EVIDENCE BASE FOR A BETTER MEASURED CURVE.
// =====
//@version=6
strategy("Gold Trend G1 - regime-scaled risk",
  overlay = true,
  initial_capital = 10000,
  currency = currency.USD,
  default_qty_type = strategy.cash,
  default_qty_value = 1000,
  commission_type = strategy.commission.cash_per_contract,
  commission_value = 0.07,
  // SLIPPAGE IS IN TICKS, NOT POINTS. XAUUSD/OANDA quotes three decimals,
  // so mintick is 0.001 and 200 ticks = 0.20 points - a realistic retail
  // gold fill. The earlier value of 5 modelled 0.005 points, roughly 40x
  // too little, and it flattered every backtest run with it.
  // COST STRESS (full period, Balanced, measured 2026-08-02):
  // 0.05 pt ( 50) -> PF 1.751 +2,533% drawdown 27.7%
  // 0.20 pt (200) -> PF 1.641 +1,620% drawdown 33.4% <- shipped
  // 0.50 pt (500) -> PF 1.452 +715% drawdown 52.2%
  // 1.00 pt (1000)-> PF 1.141 +109% drawdown 76.1%
  // 1.50 pt (1500)-> PF 0.888 -54% BREAK-EVEN IS HERE
  // Set this to YOUR broker's actual gold spread before trusting any
  // number this script reports.
  slippage = 200,
  process_orders_on_close = true,
  // BUG-012 guard. Pine's margin default of 100% silently REJECTS
  // risk-sized orders with no error and no log line. It has bitten
  // this project twice.
  margin_long = 5,
  margin_short = 5,
  calc_on_every_tick = false,
  // REQUIRED by the add-to-winner block. With Pine's default of 0,
  // strategy.entry while a position is open is silently ignored.
  pyramiding = 5,
  // The regime EMA plus its slope lookback reaches ~2,200 bars back.
  // Pine's default buffer is 300, which yields na without an error.
  max_bars_back = 5000)

// — @ Auto-scale —————
// Every lookback here is in BARS, so changing timeframe silently doubles
// or halves it. That caused two failed reproductions, so the script
// derives its own settings from the chart instead.
// 15m -> entry 48, SMA 768, EMA 2016, slope 192, trail 6.00
// 30m -> entry 24, SMA 384, EMA 1008, slope 96, trail 4.24
bool autoScale = input.bool(true, "Auto-scale settings to the chart timeframe", group="@ Auto-scale",
  tooltip="ON is strongly recommended. The inputs below are then IGNORED and derived from the timeframe, so moving
  between 15m and 30m cannot silently change the strategy. The dashboard shows what is actually in use. RUN THIS
  ON 30m: it beat 15m on profit factor, return AND drawdown in both halves.")

// — ● RISK PROFILE (the only dial most people need) —————
// Same engine, same entry, same exit. Only how HARD it presses.
// Full 6.7 years, XAUUSD 30m, gap-aware fills:
//
// profile adds risk net maxDD worst yr PF
// Conservative 3 ATR x2 1.0% +660% 17.7% -1.5% 1.626
// Balanced 1.5 ATR x4 1.0% +2,307% 29.8% -6.3% 1.728
// Aggressive 1.5 ATR x4 1.5% +8,627% 42.8% -12.1% 1.769
// Maximum 1.5 ATR x4 2.0% +24,712% 54.5% -22.5% 1.764
//
// All four are positive in all five walk-forward slices and all four lose
// only in 2021. Return roughly triples at each step and so does the pain:
// Maximum spent 2021 down 22.5% with a 45.8% intra-year drawdown. Choose
// by the drawdown you can hold through, not by the return.
string riskProf = input.string("25% DD budget", "Risk setting", options=["25% DD budget", "20% DD budget", "Custom"],
  group="● Risk Profile", tooltip="Risk per trade is SOLVED so the historical maximum drawdown lands on the
  stated budget. It is not a preference dial - the drawdown target is the input and the risk percentage is the
  output.\n\n25% DD budget: risk 0.705%. PF 1.945, net +1,875%, max DD 24.98%, MAR 2.26. Monte Carlo median DD
  25.3%, 21.5% chance of exceeding 30%.\n\n20% DD budget: risk 0.553%. PF 1.906, net +998%, max DD 20.00%, MAR
  2.17. Monte Carlo median DD 20.4%, only 5.0% chance of exceeding 30% - the most trustworthy drawdown figure
```

```

    this project has produced.\n\nA backtest drawdown is NOT an expectation. Reordering the same trades gives a
    median about 1 point worse than the historical path. Size for the Monte Carlo median, not the backtest.")

// — ● Audit findings (2026-07-31 expert pass) —————
// 35 entry x exit combinations were measured after this build shipped.
// NOTHING beat it on the criteria that predict live behaviour (zero
// losing walk-forward slices, one losing year, best return-per-drawdown
// among the zero-loss options). Two real findings came out of it:
int   coolDown = input.int(30, "Cooldown after an exit (bars)", minval=0, maxval=100, group="● Audit Findings",
    tooltip="UPDATED 2026-08-03 from 3 to 24 bars (12 hours on 30m). Re-swept 6-48 under entry-ATR add sizing:
    return/drawdown runs 64/60/72/68/74/78/84/79/58/56/22 for cooldown 6/9/12/16/20/24/28/32/36/40/48, a broad
    plateau from 12 to 32 with the baseline at 62. 24 is the middle of that plateau, not its peak (28 scores
    highest). Confirmed on US30: ret/DD 6.73 vs 4.33.\n\nORIGINAL NOTE: after an exit the shipped build could re-
    enter on the very next bar. Swept 0/3/10/24/48 bars: 3 was the free improvement - full period +2,432% to
    +2,533% and drawdown 28.55% to 27.72% with no cost anywhere else. 24 bars gives the best out-of-sample profit
    factor (2.098) but less return; 48 costs a third of the return.")
bool  longSlope = input.bool(false, "Longs ALSO require a rising EMA", group="● Audit Findings",
    tooltip="AUDIT FINDING, defaulted OFF because it is a genuine trade-off, not a free win. Shorts must have the EMA
    falling but longs were gated on nothing - an asymmetry that was never tested until now.\n\nWith longs also
    requiring a RISING EMA: profit factor 1.746 to 1.778, out-of-sample 2.065 to 2.179, and the losing year
    DISAPPEARS - 0 of 7 years negative instead of 1 of 7.\n\nThe cost: one of the five walk-forward slices turns
    negative (0/5 becomes 1/5) and drawdown rises 28.55% to 33.78%. It trades a bad year for a bad quarter. Turn
    it on if a losing calendar year is what you cannot tolerate; leave it off if drawdown is.")

// — @ Entry —————
string entryMode = input.string("Donchian breakout", "Entry trigger", options=["Donchian breakout", "HMA cross"],
    group="@ Entry",
    tooltip="Both measured through the identical exit, filters and adds.\n\nDONCHIAN: PF 1.626, +660.2%, drawdown
    17.68%, return/drawdown 37.3.\n\nHMA CROSS: PF 1.743, +540.4%, drawdown 20.9%, return/drawdown 25.9.\n\nHMA has
    the higher profit factor in BOTH halves (train 1.462 vs 1.370, out of sample 2.008 vs 1.879) - it is more
    accurate per trade. Donchian captures more and does it at less drawdown. Neither is strictly better.\n\nAlso
    measured and NOT offered: SMA and EMA crosses. The SMA 200 cross, the usual default everywhere, is the worst
    setting tested - PF 0.920 and -17.0% with no filters.")
int   entryLen = input.int(50, "Breakout lookback (bars)", minval=5, group="@ Entry",
    tooltip="Long when close exceeds the highest high of the prior N bars. Validated at 50 and 100. A timing trigger,
    not an edge.")
int   hmaLen = input.int(200, "HMA length (HMA mode only)", minval=10, group="@ Entry",
    tooltip="Swept 50/100/200/400/800; 200 best by the weaker of the two halves. NOT auto-scaled - it was measured at
    200 on 30m specifically.")
bool  longOnly = input.bool(false, "Long only (leave OFF)", group="@ Entry",
    tooltip="THIS IS A REGIME BET, NOT A SETTING. Long-only wins every rising year and loses in falling ones. In the
    Feb-Jul 2026 fall, long-only returned -5.72% live while long+short returned +7.57%.")

// — ● G1 REGIME RISK SCALING —————
// The ONLY difference from config E. Risk on a NEW ENTRY is multiplied by
// f1 x f2, clipped to [0.5, 2.0]:
// f1 = 1.4 when |close - regime EMA| / ATR > 6.0, else 0.8
// f2 = 1.4 when SMA(ATR,20) / SMA(ATR,200) > 1.05, else 0.6
// i.e. press when the market is BOTH far from its mean AND expanding;
// shrink when it is neither. Both inputs are causal - no lookahead.
bool  useRegimeRisk = input.bool(true, "G1: scale risk by trend strength x volatility expansion", group="● G1 Regime
    Risk")
float trendThr = input.float(6.0, "Trend-strength threshold (ATR from the regime EMA)", minval=0.5, step=0.5,
    group="● G1 Regime Risk")
float expThr = input.float(1.05, "Volatility-expansion threshold (fast ATR / slow ATR)", minval=0.5, step=0.01,
    group="● G1 Regime Risk")
float upMult = input.float(1.4, "Multiplier when the condition holds", minval=1.0, maxval=2.0, step=0.05, group="●
    G1 Regime Risk")
float dnMultT = input.float(0.8, "Trend multiplier when it does not", minval=0.3, maxval=1.0, step=0.05, group="●
    G1 Regime Risk")
float dnMultV = input.float(0.6, "Volatility multiplier when it does not", minval=0.3, maxval=1.0, step=0.05,
    group="● G1 Regime Risk")

// — @ Risk and filters —————
float riskPct = input.float(1.0, "Risk per trade (%)", step=0.25, group="@ Risk & Filters",
    tooltip="The direct return lever, and it is not free. Measured: 1% -> +85.4% train at 13.6% drawdown; 2% ->
    +208.8% at 28.6%; 3% -> +367.4% at 41.8%. Return scales roughly linearly and so does drawdown. Choose by the
    drawdown you can sit through.")
float maxLev = input.float(20, "Max leverage (notional cap)", minval=1, group="@ Risk & Filters",
    tooltip="BUG-012 guard. Without a notional cap, risk-based sizing can request more than buying power and
    TradingView skips the order with no error.")
float trailMult = input.float(6.0, "Trailing stop (x ATR, 15m basis)", minval=1.0, step=0.5, group="@ Risk & Filters",
    tooltip="THE key parameter, and the one whose old description was wrong. Wider raises PROFIT FACTOR but LOWERS
    RETURN, because risk-based sizing shrinks the position as the stop widens: trail 20 ATR gives PF 1.659 but
    only +23.5%, against trail 6's 1.243 and +85.4%. 6 on a 15m basis (4.24 on 30m) is return-optimal at tolerable
    drawdown.")
int   atrLen = input.int(14, "ATR length", minval=2, group="@ Risk & Filters")
bool  useSma = input.bool(true, "Fast SMA must agree", group="@ Risk & Filters",
    tooltip="750 bars on 15m, about 8 days. Selected on the training half only.")
int   smaLen = input.int(750, "Fast SMA length (bars)", minval=50, group="@ Risk & Filters")
bool  useSma2 = input.bool(true, "Slow SMA must agree TOO (the confluence layer)", group="@ Risk & Filters",
    tooltip="Requiring BOTH averages lifts full-period PF 1.379 -> 1.421 and cuts drawdown 12.30% -> 9.63%. Return
    barely moved; the drawdown fell by a fifth. It is a risk filter, not a return generator.\n\nIt is the
    AGREEMENT that works, not the slower average: SMA2000 instead of SMA750 is worse (train 1.181 vs 1.289), and
    requiring only ONE of the two is much worse (1.149). The two horizons disagreeing marks the choppy middle of a
    range, and that is what gets skipped.")
int   sma2Len = input.int(2000, "Slow SMA length (bars)", minval=100, group="@ Risk & Filters")
bool  useRegime = input.bool(true, "EMA regime gate", group="@ Risk & Filters",
    tooltip="Only long above the EMA, only short below it.")
int   regimeLen = input.int(2000, "Regime EMA length (bars)", minval=20, group="@ Risk & Filters")
bool  useSlope = input.bool(true, "Shorts only when the EMA is FALLING", group="@ Risk & Filters",
    tooltip="This is what makes the system profitable in a falling market without giving up the trend returns. Price
    sits below a FLAT EMA constantly during uptrend pullbacks; the EMA actually declining is far rarer and is what
    a bear regime looks like. Gating shorts on slope keeps them nearly silent in bull markets, where counter-
    trend shorts are what bleed away the edge.")
int   slopeLen = input.int(200, "Slope lookback (bars)", minval=20, group="@ Risk & Filters")
float shortRisk = input.float(0.75, "Short risk multiplier", minval=0.1, maxval=3.0, step=0.05, group="@ Risk &
    Filters",
    tooltip="Counter-trend trades sized smaller. 0.75 gave the best full-period result; raise toward 1.5 for more
    downside protection at the cost of total return.")

// — @ Add to winners —————
bool  usePyr = input.bool(true, "Add to a winning position", group="@ Add to Winners",
    tooltip="THE BIGGEST RETURN LEVER MEASURED HERE, and it multiplies BOTH sides. Walk-forward over five consecutive
    slices, net/drawdown, without adds vs with 2 adds every 3 ATR:\n\n+38.4/9.2 -> +92.5/16.3\n+19.6/6.7

```

```

-> +23.9/13.5\n +8.8/10.4 -> +11.9/16.6\n +15.1/10.2 -> +33.3/16.8\n +28.4/7.8 -> +88.1/13.5\n\nEvery
slice improves, which is why it ships on. But through RANDOM entries it lifts the median from 6.8% to 25.3%
while taking the spread from 6.8 to 29.5 and the worst seed from -3.5% to -30.2%. It is not an edge. It is
betting more on trades that are already working.")
float pyrAtr = input.float(3.0, "Add every (x ATR of favourable move)", minval=0.5, step=0.5, group="Ⓢ Add to
Winners",
tooltip="3.0 with 2 adds is the moderate setting. 2.0 with 4 adds returns far more (+163% and +168% on the first
and last slices) at drawdowns up to 26.4%.")
int pyrMax = input.int(2, "Maximum adds", minval=1, maxval=4, group="Ⓢ Add to Winners")
float pyrRisk = input.float(1.0, "Each add sized at (x original risk)", minval=0.1, maxval=1.0, step=0.1, group="Ⓢ
Add to Winners")

// — Ⓢ Trailing algorithm
string trailMode= input.string("Chandelier (ATR)", "Trailing logic", options=["Chandelier (ATR)", "Donchian low"],
group="Ⓢ Trailing Algorithm",
tooltip="Six algorithms were built and measured: chandelier, Donchian low, EMA, a multiple that tightens with
profit, a give-back-a-fraction trail, and none. Two are worth offering.\n\nCHANDELIER (default): PF 1.635,
+679.9%, drawdown 17.01%, positive in ALL FIVE walk-forward slices.\nDONCHIAN LOW: PF 1.624, +935.9%, drawdown
23.45%, NEGATIVE in one of five (-6.0%).\n\nDonchian returns 38% more, but return per drawdown is a dead heat
(39.9 vs 40.0) - the same efficiency at larger size, with one losing period to sit through.\n\nRejected: EMA
trail (best PF 1.383 at 33.7% drawdown), give-back trail (1.061), stepping trail (1.597, behind both).")
float tightAfter= input.float(15.0, "Tighten the trail after this much favourable move (x ATR at entry)", minval=0,
step=1.0, group="Ⓢ Trailing Algorithm",
tooltip="0 disables. PRIORITY 4 FINDING (2026-08-03). Once a trade has run this far in its favour, the leash
tightens to the value below. This is NOT a tight stop and NOT a profit target: it only ever engages on trades
already deep in profit, and the trade still exits on a trail.\n\nAt 20 ATR -> 2.0: profit factor 1.714 ->
1.834, net +3,855% -> +5,922%, drawdown 32.65% -> 33.80%. Against a leverage-matched baseline at the same
drawdown it returns +5,922% vs about +4,500% with a higher profit factor, so it is not the risk dial in
disguise. Improves the profit factor in 5 of 8 calendar years. Confirmed on US30: PF 1.469 vs 1.351 at
identical drawdown.\n\nSwept 10-30 ATR x 1.5-3.0: 21 of 28 combinations beat the baseline and everything from
18 ATR up is a plateau. 20 is inside it, not its peak.")
float tightTo = input.float(2.0, "...tighten it to (x ATR)", minval=0.5, step=0.25, group="Ⓢ Trailing Algorithm")
int donchN = input.int(45, "Donchian trail lookback (bars)", minval=5, group="Ⓢ Trailing Algorithm",
tooltip="The stop rides the lowest low of the last N bars for longs. Do NOT read the training peak at 40 as
meaningful: out of sample the whole 30-50 range performs and the ordering reverses, which is what a fitted
spike looks like once you test it. 45 is the middle of the out-of-sample plateau, not the training maximum.")

// — Ⓢ TP1 / TP2 (staged profit taking, OFF by default)
bool tp1On = input.bool(false, "TP1 - bank part of the position", group="Ⓢ TP1 / TP2",
tooltip="MEASURED 20 WAYS AND IT COSTS RETURN EVERY TIME. Best variant: +234.1% against +253.1% with no target.
Adding TP2 as well: +216.7%. Out of sample the gap widens - PF 1.568 with TP1 against 1.930 without. Even
restricting targets to the counter-trend side fails (+242.6%), the only place in this project where that
asymmetry rule has not held.\n\nWhat it DOES buy is a smoother curve: full-period drawdown 17.01% -> 16.04%,
out-of-sample drawdown 12.79% -> 8.38%. A real trade, just not a free one.")
float tp1Atr = input.float(4.0, "TP1 distance (x ATR)", minval=0.5, step=0.5, group="Ⓢ TP1 / TP2")
float tp1Pct = input.float(25, "TP1 - % of position closed", minval=5, maxval=90, step=5, group="Ⓢ TP1 / TP2")
bool tp2On = input.bool(false, "TP2 - bank another part", group="Ⓢ TP1 / TP2")
float tp2Atr = input.float(12.0, "TP2 distance (x ATR)", minval=1.0, step=0.5, group="Ⓢ TP1 / TP2")
float tp2Pct = input.float(25, "TP2 - % of position closed", minval=5, maxval=90, step=5, group="Ⓢ TP1 / TP2")
bool tpShorts = input.bool(false, "Apply TP1/TP2 to shorts only", group="Ⓢ TP1 / TP2")

// — Ⓢ Pure profit-taking mode (different architecture)
string tpMode = input.string("Off", "Fixed-target mode", options=["Off", "Percent of price", "Fixed points"],
group="Ⓢ Profit-Taking Mode",
tooltip="TURNS THIS INTO A PROFIT-TAKING STRATEGY instead of a trend-following one. The trail and the adds are
bypassed: entry, static stop, fixed target, done.\n\nMeasured with a 4 ATR stop and a 4% target: train PF
1.402 / +98.5%, out of sample PF 1.420 / +23.2%, full period PF 1.475 / +185.3% at 23.2% drawdown. Train and
out of sample agree to 0.018 of profit factor - the most stable result in this project.\n\nIt beats the
trailing build on profit factor (1.475 vs 1.444 without adds) and loses badly on drawdown and return. Use it
if you want frequent closed wins; leave it Off if you want the return.\n\nUSE PERCENT, NOT POINTS: gold ran
1450 -> 4100 over the sample, so a fixed 40-point target was 2.7% of price at the start and 1.0% at the end.
Percent scored PF 1.475 against points' 1.160.")
float tpPct = input.float(4.0, "Target - percent of price", minval=0.25, step=0.25, group="Ⓢ Profit-Taking Mode",
tooltip="Swept 1.5/2/3/4%. Bigger is monotonically better on profit factor; 4% is where drawdown stops
improving.")
float tpPoints = input.float(160, "Target - fixed points", minval=1, step=5, group="Ⓢ Profit-Taking Mode")
float tpStopAtr = input.float(4.0, "Static stop for this mode (x ATR)", minval=0.5, step=0.5, group="Ⓢ Profit-Taking
Mode",
tooltip="Swept 2/3/4/6/8 ATR. 4 was best; 2 was catastrophic (43% drawdown) because the target then needs an
unreachable hit rate.")

// — Ⓢ Display
bool showStop = input.bool(true, "Plot the stop", group="Ⓢ Display")
bool showBand = input.bool(true, "Plot the entry trigger", group="Ⓢ Display")
bool showDash = input.bool(true, "Show dashboard", group="Ⓢ Display")

// =====
// EFFECTIVE SETTINGS
// Targets: breakout ~12h, fast SMA ~8 days, EMA ~21 days, slope ~2 days.
// The trail scales with the SQUARE ROOT of the timeframe because ATR
// grows roughly as sqrt(time); that rule was validated by sweeping the
// 30m trail directly (4.24 was best on train net, +113.5%).
// =====
// Profile presets override the manual risk and add settings.
float riskE = riskProf == "20% DD budget" ? 0.567 : riskProf == "Custom" ? riskPct : 0.723
float pyrAtrE = riskProf == "Custom" ? pyrAtr : 1.5
int pyrMaxE = riskProf == "Custom" ? pyrMax : 4

float tfMin = timeframe.in_seconds() / 60.0
int entryLenE = autoScale ? math.max(5, int(12.0 * 60.0 / tfMin)) : entryLen
int smaLenE = autoScale ? math.max(50, int(8.0 * 24.0 * 60.0 / tfMin)) : smaLen
int regimeLenE = autoScale ? math.max(20, int(21.0 * 24.0 * 60.0 / tfMin)) : regimeLen
int sma2LenE = autoScale ? math.max(100, int(13.0 * 24.0 * 60.0 / tfMin)) : sma2Len
int slopeLenE = autoScale ? math.max(20, int(2.0 * 24.0 * 60.0 / tfMin)) : slopeLen
float trailMultE = autoScale ? 6.0 * math.sqrt(15.0 / tfMin) : trailMult

// =====
// SIGNALS - every ta.* call unconditional at top level (BUG-004)
// =====
float atr = ta.atr(atrLen)
float donHi = ta.highest(high, entryLenE)[1]
float donLo = ta.lowest(low, entryLenE)[1]
float donLoTrail = ta.lowest(low, donchN)[1]
float donHiTrail = ta.highest(high, donchN)[1]
float regimeMa = ta.ema(close, regimeLenE)

```

```

float trendSma    = ta.sma(close, smaLenE)
float slowSma     = ta.sma(close, sma2LenE)
float hmaVal      = ta.hma(close, hmaLen)
bool  hmaUp       = ta.crossover(close, hmaVal)
bool  hmaDn       = ta.crossunder(close, hmaVal)

// G1 regime multiplier. Uses only closed-bar information.
float trendStr    = atr > 0 ? math.abs(close - regimeMa) / atr : 0.0
float atrFast     = ta.sma(atr, 20)
float atrSlow     = ta.sma(atr, 200)
float volExp      = atrSlow > 0 ? atrFast / atrSlow : 1.0
float gif1        = trendStr > trendThr ? upMult : dnMultT
float gif2        = volExp > expThr ? upMult : dnMultV
float regimeMult  = useRegimeRisk ? math.min(math.max(gif1 * gif2, 0.5), 2.0) : 1.0
bool  emaFalling  = regimeMa < regimeMa[slopeLenE]
bool  emaRising   = regimeMa > regimeMa[slopeLenE]
bool  regimeOkL   = (not useRegime or close > regimeMa) and (not longSlope or emaRising)
bool  regimeOkS   = (not useRegime or close < regimeMa) and (not useSlope or emaFalling)
bool  smaOkL      = not useSma or (not na(trendSma) and close > trendSma)
bool  smaOkS      = not useSma or (not na(trendSma) and close < trendSma)
bool  sma2OkL     = not useSma2 or (not na(slowSma) and close > slowSma)
bool  sma2OkS     = not useSma2 or (not na(slowSma) and close < slowSma)

bool  trigUp      = entryMode == "HMA cross" ? hmaUp : (not na(donHi) and close > donHi)
bool  trigDn      = entryMode == "HMA cross" ? hmaDn : (not na(donLo) and close < donLo)

bool  longSig     = trigUp and regimeOkL and smaOkL and sma2OkL
bool  shortSig    = not longOnly and trigDn and regimeOkS and smaOkS and sma2OkS

// =====
// STATE
// =====
bool  tpOn = tpMode != "Off"

var int barsSinceExit = 0
barsSinceExit := strategy.position_size == 0 ? barsSinceExit + 1 : 0
bool  coolOk = barsSinceExit > coolDown

var float trailStop = na
var float tpStop    = na
var float tpTarget  = na
var float tp1Px     = na
var float tp2Px     = na
var bool  tp1Done   = false
var bool  tp2Done   = false
var float lastAdd   = na
var int   nAdds     = 0
// ATR AT ENTRY, for add sizing. PRIORITY 2 FINDING (2026-08-03): sizing each
// add off the CURRENT ATR shrinks it exactly when volatility is expanding,
// which on this system is precisely when the trade is working. Sizing off the
// entry ATR - equivalently, every add equals the ORIGINAL position size -
// scored PF 1.647 vs 1.623, net +1,996% vs +1,617% and drawdown 32.2% vs
// 36.5%. Against a leverage-matched baseline at the same drawdown it returned
// +1,996% vs +1,094% with a higher PF, so it is not the risk dial in disguise.
var float atrEntry  = na
var float bestPx    = na // best excursion THROUGH THE PREVIOUS BAR (see below)

// =====
// TRAILING STOP
// Ratchets in the trade's favour only, anchored to the PREVIOUS bar's
// extreme so the current bar can never move its own stop and then hit it.
// =====
// Excursion is measured off bestPx, which holds the extreme through the
// PREVIOUS bar only. Using this bar's own high would let the bar tighten its
// own stop and then hit it -- the intrabar-lookahead defect of BUG-019.
float excAtr = strategy.position_size == 0 or na(atrEntry) or na(bestPx) or atrEntry <= 0 ? 0.0 :
    (strategy.position_size > 0 ? bestPx - strategy.position_avg_price
     : strategy.position_avg_price - bestPx) / atrEntry
float effTrail = tightAfter > 0 and excAtr >= tightAfter ? tightTo : trailMultE

if strategy.position_size > 0
    float cand = trailMode == "Donchian low" ? donLoTrail : high[1] - atr * effTrail
    trailStop := na(trailStop) ? cand : math.max(trailStop, cand)
else if strategy.position_size < 0
    float cand = trailMode == "Donchian low" ? donHiTrail : low[1] + atr * effTrail
    trailStop := na(trailStop) ? cand : math.min(trailStop, cand)
else
    trailStop := na
    lastAdd   := na
    atrEntry  := na
    bestPx    := na
    nAdds     := 0

// =====
// ADD TO A WINNING POSITION
// PORT DELTA: with process_orders_on_close the add fills at THIS bar's
// close, so the enlarged position only meets the trailing stop from the
// next bar. The research engine adds at the trigger price and exposes the
// enlarged size on the same bar, so Python is the stricter of the two.
// =====
if usePyr and not tpOn and strategy.position_size != 0 and nAdds < pyrMaxE and not na(lastAdd)
    float step = atr * pyrAtrE
    bool  isLong = strategy.position_size > 0
    float nxt    = isLong ? lastAdd + step : lastAdd - step
    if isLong ? high >= nxt : low <= nxt
        float aRisk = (isLong ? riskE : riskE * shortRisk) * pyrRisk
        float aAtr  = na(atrEntry) ? atr : atrEntry // entry-ATR sizing
        float aQty  = math.floor(math.min(strategy.equity * aRisk / 100.0 / (aAtr * trailMultE),
            strategy.equity * maxLev / close - math.abs(strategy.position_size)))
        if aQty >= 1
            strategy.entry(isLong ? "L" : "S", isLong ? strategy.long : strategy.short, qty=aQty)
            nAdds := nAdds + 1
            lastAdd := nxt

```

```

// ENTRIES
//
bool wantShort = shortSig and not longSig
// NOTE: the multiplier is applied to the OPENING position only. Pyramid adds
// stay at base risk. That asymmetry is what was tested and measured; scaling
// the adds too was not, so it is not shipped.
float effRisk = (wantShort ? riskE * shortRisk : riskE) * regimeMult
float stopDist = atr * trailMultE
float qtyRaw = stopDist > 0 ? strategy.equity * effRisk / 100.0 / stopDist : 0
float qtyCap = strategy.equity * maxLev / close
float qty = math.floor(math.min(qtyRaw, qtyCap))

if longSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("L", strategy.long, qty=qty)
    trailStop := close - stopDist
    tpStop := close - atr * tpStopAtr
    tpTarget := tpMode == "Fixed points" ? close + tpPoints : close * (1.0 + tpPct / 100.0)
    tp1Px := close + atr * tp1Atr
    tp2Px := close + atr * tp2Atr
    tp1Done := false
    tp2Done := false
    lastAdd := close
    atrEntry := atr
    bestPx := close
    nAdds := 0

if shortSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("S", strategy.short, qty=qty)
    trailStop := close + stopDist
    tpStop := close + atr * tpStopAtr
    tpTarget := tpMode == "Fixed points" ? close - tpPoints : close * (1.0 - tpPct / 100.0)
    tp1Px := close - atr * tp1Atr
    tp2Px := close - atr * tp2Atr
    tp1Done := false
    tp2Done := false
    lastAdd := close
    atrEntry := atr
    bestPx := close
    nAdds := 0

//
// STAGED PROFIT TAKING (off by default)
// PORT DELTA: these fill at the bar CLOSE via strategy.close, not at the
// target price. Two live limit orders plus a trailing stop on one entry
// is where order conflicts come from. The research engine fills at the
// target, so expect Pine to read slightly worse here.
//
bool tpSideOk = not tpShorts or strategy.position_size < 0
bool tp1Hit = tp1On and not tpOn and not tp1Done and tpSideOk and strategy.position_size != 0 and
    (strategy.position_size > 0 ? high >= tp1Px : low <= tp1Px)
bool tp2Hit = tp2On and not tpOn and not tp2Done and tpSideOk and strategy.position_size != 0 and
    (strategy.position_size > 0 ? high >= tp2Px : low <= tp2Px)

if tp1Hit
    strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp1Pct, comment="TP1")
    tp1Done := true
if tp2Hit
    strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp2Pct, comment="TP2")
    tp2Done := true

//
// EXITS
// Fixed-target mode replaces the ratchet entirely: static stop, fixed
// limit, no trail and no adds. Anything else would be a third
// architecture that was never measured.
//
if tpOn
    if strategy.position_size > 0 and not na(tpStop)
        strategy.exit("L-exit", "L", stop=tpStop, limit=tpTarget)
    if strategy.position_size < 0 and not na(tpStop)
        strategy.exit("S-exit", "S", stop=tpStop, limit=tpTarget)
else
    if strategy.position_size > 0 and not na(trailStop)
        strategy.exit("L-exit", "L", stop=trailStop)
    if strategy.position_size < 0 and not na(trailStop)
        strategy.exit("S-exit", "S", stop=trailStop)

// Roll bestPx forward only now that this bar's stop and exits are resolved,
// so the value the next bar reads is the extreme through THIS bar and no
// bar can ever act on its own high.
if strategy.position_size != 0
    bestPx := na(bestPx) ? close : (strategy.position_size > 0 ? math.max(bestPx, high) : math.min(bestPx, low))

//
// VISUALS
//
plot(showBand and entryMode == "Donchian breakout" and not na(donHi) ? donHi : na, "Breakout high",
    color.new(color.orange, 40), 1, plot.style_stepline)
plot(showBand and entryMode == "Donchian breakout" and not na(donLo) ? donLo : na, "Breakout low",
    color.new(color.orange, 70), 1, plot.style_stepline)
plot(showBand and entryMode == "HMA cross" ? hmaVal : na, "HMA", color.new(color.fuchsia, 0), 2)
plot(showStop and strategy.position_size != 0 ? (tpOn ? tpStop : trailStop) : na, "Stop", color.new(#D50000, 0), 2,
    plot.style_stepline)
plot(tpOn and strategy.position_size != 0 ? tpTarget : na, "Fixed target", color.new(#00E676, 0), 2,
    plot.style_stepline)
plot(tp1On and not tpOn and strategy.position_size != 0 and not tp1Done ? tp1Px : na, "TP1", color.new(#00E676, 20),
    1, plot.style_circles)
plot(tp2On and not tpOn and strategy.position_size != 0 and not tp2Done ? tp2Px : na, "TP2", color.new(#00E676, 45),
    1, plot.style_circles)
plot(useRegime ? regimeMa : na, "Regime EMA", color.new(color.fuchsia, 55), 1)
plot(useSma ? trendSma : na, "Fast SMA", color.new(color.aqua, 25), 2)
plot(useSma2 ? slowSma : na, "Slow SMA", color.new(#7E57C2, 15), 2)

plotshape(longSig and strategy.position_size == 0, "Long entry", shape.triangleup, location.belowbar,
    color.new(#00C853, 0), size=size.small, text="LONG")

```

```

plotshape(shortSig and strategy.position_size == 0, "Short entry", shape.triangledown, location.abovebar,
color.new(#D50000, 0), size=size.small, text="SHORT")

bool exitedLong = strategy.position_size[1] > 0 and strategy.position_size == 0
bool exitedShort = strategy.position_size[1] < 0 and strategy.position_size == 0
plotshape(exitedLong or exitedShort, "Exit", shape.xcross, location.absolute, color.new(color.gray, 0),
size=size.tiny)

// =====
// ALERTS
// =====
alertcondition(longSig, "Gold Trend - LONG entry", "LONG {{ticker}} {{interval}} at {{close}}")
alertcondition(shortSig, "Gold Trend - SHORT entry", "SHORT {{ticker}} {{interval}} at {{close}}")
alertcondition(exitedLong or exitedShort, "Gold Trend - position closed", "EXIT {{ticker}} {{interval}} at {{close}}")
alertcondition(tp1Hit, "Gold Trend - TP1 hit", "TP1 {{ticker}} {{interval}} at {{close}}")
alertcondition(tp2Hit, "Gold Trend - TP2 hit", "TP2 {{ticker}} {{interval}} at {{close}}")

// =====
// DASHBOARD
// =====
var table dash = table.new(position.top_right, 2, 10, bgcolor=color.new(color.black, 20), border_width=1,
border_color=color.new(color.gray, 60))
if showDash and barstate.islast
int nT = strategy.closedtrades
int wT = 0
float gw = 0.0
float gl = 0.0
if nT > 0
for i = 0 to nT - 1
float p = strategy.closedtrades.profit(i)
wT += p > 0 ? 1 : 0
gw += p > 0 ? p : 0.0
gl += p < 0 ? -p : 0.0
float wr = nT > 0 ? wT * 100.0 / nT : na
float pf = gl > 0 ? gw / gl : na
float netPct = (strategy.equity - strategy.initial_capital) / strategy.initial_capital * 100
int nUp = (regimeOkL ? 1 : 0) + (smaOkL ? 1 : 0) + (sma2OkL ? 1 : 0)
int nDn = (regimeOkS ? 1 : 0) + (smaOkS ? 1 : 0) + (sma2OkS ? 1 : 0)

table.cell(dash, 0, 0, "GOLD TREND", text_color=color.orange, text_size=size.small)
table.cell(dash, 1, 0, riskProf + " " + str.tostring(riskE, "#.#") + "%", text_color=color.orange,
text_size=size.small)
table.cell(dash, 0, 1, "Trades", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 1, str.tostring(nT), text_color=nT < 250 ? color.yellow : color.white, text_size=size.tiny)
table.cell(dash, 0, 2, "Win rate", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 2, na(wr) ? "-" : str.tostring(wr, "#.#") + "%", text_color=color.white, text_size=size.tiny)
table.cell(dash, 0, 3, "Profit factor", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 3, na(pf) ? "-" : str.tostring(pf, "#.##"), text_color=na(pf) ? color.gray : pf >= 1.3 ?
color.lime : pf >= 1.0 ? color.yellow : color.red, text_size=size.tiny)
table.cell(dash, 0, 4, "Net", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 4, str.tostring(netPct, "#.#") + "%", text_color=netPct > 0 ? color.lime : color.red,
text_size=size.tiny)
table.cell(dash, 0, 5, "Position", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 5, strategy.position_size > 0 ? "LONG " + str.tostring(nAdds) : strategy.position_size < 0 ?
"SHORT " + str.tostring(nAdds) : "flat", text_color=strategy.position_size != 0 ? color.lime : color.gray,
text_size=size.tiny)
table.cell(dash, 0, 6, "Stop", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 6, na(trailStop) and na(tpStop) ? "-" : str.tostring(tpOn ? tpStop : trailStop,
format.mintick), text_color=color.red, text_size=size.tiny)
table.cell(dash, 0, 7, "Confluence", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 7, nUp == 3 ? "LONG 3/3" : nDn == 3 ? "SHORT 3/3" : "none (" + str.tostring(math.max(nUp,
nDn)) + "/3)", text_color=nUp == 3 ? color.lime : nDn == 3 ? color.red : color.gray, text_size=size.tiny)
table.cell(dash, 0, 8, "Exit engine", text_color=color.silver, text_size=size.tiny)
table.cell(dash, 1, 8, tpOn ? "fixed target" : trailMode == "Donchian low" ? "donchian " + str.tostring(donchN) :
"chandelier " + str.tostring(trailMultE, "#.##"), text_color=color.aqua, text_size=size.tiny)
table.cell(dash, 0, 9, autoScale ? "auto " + timeframe.period : "MANUAL", text_color=autoScale ? color.aqua :
color.orange, text_size=size.tiny)
table.cell(dash, 1, 9, autoScale ? "sma " + str.tostring(smaLenE) + "/" + str.tostring(sma2LenE) : (nT < 250 ?
"n<250: unstable" : ""), text_color=color.silver, text_size=size.tiny)

```

— end of Pine source (480 lines) —

Version 4 — Config G1* (CURRENT SHIPPED)

1. Identity and provenance

Field	Value
Label in research	Config G1* ("G1 star")
Pine file	strategies/gold_trend_G1.pine (current HEAD)
Commit	7b25357 — "adopt G1*: regime multiplier extended to pyramid adds"
Adoption date	2026-08-05
Change vs previous	The G1 regime multiplier, captured at entry, now also scales every pyramid add.
Was it shipped/live?	YES — this is the current shipped champion.

2. Complete rule specification

2.1 Entry

Field	Value
Bar close only	Yes. Every condition is evaluated on the CLOSED bar. No intrabar evaluation, no varip, no request.security with loc
Donchian breakout lookback	24 bars = 12.0 hours on 30m. Pine: entryLenE = max(5, int(12.0*60/tfMin)). Applied with a [1] offset: donHi = ta.hig
Regime EMA	EMA(close, 1008 bars) = 21.0 days on 30m. Pine: regimeLenE = max(100, int(21.0*24*60/tfMin)). Bar-based, not c
Fast SMA	SMA(close, 384 bars) = 8.0 days on 30m. Pine: smaLenE = max(50, int(8.0*24*60/tfMin)).
Slow SMA	SMA(close, 630 bars) = 13.0 days
Slope lookback	96 bars = 2.0 days on 30m. Pine: slopeLenE = max(20, int(2.0*24*60/tfMin)).
LONG conditions (all required)	1) close > ta.highest(high,24)[1] 2) close > EMA(close,1008) 3) close > SMA(close,384) 4) close > SMA(close,S
SHORT conditions (all required)	1) close < ta.lowest(low,24)[1] 2) close < EMA(close,1008) 3) close < SMA(close,384) 4) close < SMA(close,SL
Other entry filters	NONE. No session filter, no volume filter, no volatility filter, no candlestick or level confluence.
Cooldown after an exit	30 bars — a new entry requires barsSinceExit > cooldown.

2.2 Exit

Field	Value
Initial trail multiple	4.24 x ATR(14) initially, tightening to 2.0.
Number of exits	Exactly one: a ratcheting chandelier trailing stop.
ATR period	ATR(14), Wilder smoothing (RMA / ewm alpha=1/14).
Tightening rule	Identical to E and G1: 4.24 -> 2.0 x ATR after 15 x ATR(entry).
Trail anchor	PREVIOUS bar's extreme. Long: high[1] - atr*mult. Short: low[1] + atr*mult. A bar can never move its own stop and
Ratchet	Monotone in the trade's favour only. Long: trailStop := max(trailStop, cand). Short: min(trailStop, cand).
Profit target	NONE.
Time exit	NONE.
Breakeven stop	NONE.
Max holding period	Unbounded.

2.3 Pyramiding

Field	Value
Maximum adds	4
Add trigger spacing	every 1.5 x ATR of favourable movement
Add size formula	aQty = equity * (riskE * shortRiskIfShort * pyrRisk * rmEntry) / 100 / (ATR_AT_ENTRY * 4.24)
Regime multiplier on adds	YES. rmEntry is the regimeMult value captured at ENTRY and held for the life of the trade (Pine: var float rmEntry,
Open risk treatment	Unbounded by design. Mean adds per trade is 1.809, IDENTICAL to G1 — the trade paths are the same, only the a

2.4 Position sizing and risk

Field	Value
Risk base	strategy.equity
Size formula	qty = floor(min(equity * effRisk / 100 / stopDist, equity * 20 / close)) where effRisk = (wantShort ? riskE * 0.75 : riskE

Field	Value
stopDist	ATR(14) * 4.24
Short risk multiplier	0.75
Regime multiplier — trend factor f1	trendStr = close - EMA(close,1008) / ATR(14). f1 = 1.4 if trendStr > 6.00 else 0.8
Regime multiplier — volatility factor f2	volExp = SMA(ATR(14),20) / SMA(ATR(14),200). f2 = 1.4 if volExp > 1.05 else 0.6
Regime multiplier — combination	regimeMult = clip(f1 * f2, 0.5, 2.0)
Multiplier persistence	rmEntry := regimeMult at entry; rmEntry := na when flat
Risk solved to 25% DD budget	0.646% (Pine input default 0.606 — see note)
Risk solved to 20% DD budget	0.506% (Pine input default 0.475 — see note)

2.5 Costs and execution assumptions used in research

Field	Value
Commission	USD 0.07 per contract (ounce) per side
Slippage	0.20 price points per side, applied against the trade
Notional cap	20x equity (maxLev). qty is capped at equity*20/close.
Overnight financing	NOT MODELLED. At ~38.7% time in market and ~2.6x time-weighted leverage, financing at 2-8% annual is an esti
Fill model	Entry at the signal bar's close. Stops are gap-aware (filled at the open if the bar gapped through). A bar touching bo

3. Full performance record

Gold XAUUSD 30m, 78,695 bars, 2019-12-01 to 2026-07-30 (6.66 years). Costs applied. Risk% is the SOLVED value that produced the stated drawdown.

Metric	25% DD budget	20% DD budget
Risk per trade (solved)	0.646%	0.506%
Trades (total)	711	711
Trades long	467	467
Trades short	244	244
Win rate	21.66%	21.66%
Average R	1.663	1.365
Profit factor	2.157	2.116
Net return	2,463.7%	1,246.5%
CAGR	62.78%	47.77%
Max drawdown (historical)	25.00%	20.00%
MAR (CAGR / MaxDD)	2.511	2.389
Avg hold (30m bars)	40.7	40.7
Avg hold (days)	0.85	0.85
Avg adds per trade	1.809	1.809
Avg winning trade (points)	32.77	32.77
Avg losing trade (points)	13.58	13.58
Monte Carlo median DD	21.12%	16.87%
Monte Carlo 95th pct DD	30.93%	25.02%
P(DD > 30%)	6.48%	0.85%
P(DD > 30%) std dev across seeds	0.176	0.077
Profit factor, pre-2025	1.600	1.626
PF TRAIN (first third)	1.712	1.773
PF VALID (middle third)	1.492	1.498
PF TEST (final third)	2.474	2.488
PF first half	1.446	1.493
PF second half	2.378	2.373
Top 5 trades, % of NET profit	42.3%	36.5%
Top 10 trades, % of NET profit	70.7%	61.5%
Top 5 trades, % of GROSS profit	22.7%	19.3%

Metric	25% DD budget	20% DD budget
Top 10 trades, % of GROSS profit	37.9%	32.4%

3.1 Year by year (25% drawdown budget)

Year	2019	2020	2021	2022	2023	2024	2025	2026
Trades	3	96	114	112	112	107	108	59
Profit factor	24.515	2.625	0.827	1.259	1.787	1.691	2.554	2.677

Down/flat years in this sample are 2021 (gold -4.3%), 2022 (-0.3%) and 2026 (-6.0%).

3.2 Anchored walk-forward, 6 folds (25% drawdown budget)

Fold	1	2	3	4	5	6
Profit factor	0.925	2.409	1.451	2.028	2.885	2.474

3.3 Reference: same configuration at a flat 1.0% risk

Risk	Trades	Profit factor	Net return	Max DD
1.000%	712	2.218	9,899.5%	36.59%

5. Differences from the previous version

- Regime multiplier now applies to pyramid adds as well as the opening unit, using the value captured AT ENTRY.
- New Pine state variable: var float rmEntry, assigned at entry, reset on flat.
- Add-size line changed to: $aRisk = (isLong ? riskE : riskE * shortRisk) * pyrRisk * nz(rmEntry, 1.0)$
- Risk preset re-solved downward: 0.723 -> 0.606 (25% budget), 0.567 -> 0.475 (20%).
- Entry, exit, trail, tightening, add count, add spacing, add-ATR basis and the multiplier formula itself are UNCHANGED from G1.

4. Complete Pine Script — Version 4 — Config G1* (CURRENT SHIPPED)

Source: strategies/gold_trend_G1.pine. Reproduced in full with no elisions. Inputs carry the exact defaults that reproduce the configuration documented above. Closed-bar only.

```
// =====
// GOLD TREND - CONFIG G1 (config E + regime-scaled entry risk)
// =====
// Config E in every respect except one: the risk on a NEW ENTRY is scaled
// by how far price sits from its regime mean AND whether volatility is
// expanding. Press in strong expanding trends, shrink otherwise.
//
// XAUUSD 30m, Dec 2019 - Jul 2026, 0.20 pt slippage/side, 711 trades
// budget risk PF net maxDD MAR MC med P(DD>30%)
// 25% 0.723% 2.005 +2,457% 24.99% 2.51 24.8% 19.0%
// 20% 0.567% 1.967 +1,250% 20.00% 2.39 20.0% 5.0%
//
// Against config E it is better on EVERY metric on gold at both budgets,
// including a LOWER Monte Carlo median and tail - improving the drawdown
// distribution, not just the single historical path.
//
// WHY IT IS BELIEVED
// The same effect appears through three unrelated definitions (trend
// distance, ATR expansion ratio, ATR percentile) and is a plateau, not a
// spike: trend thresholds 3/4/5/6/7/8/10 ATR all beat config E.
//
// THE QUALIFIER, STATED PLAINLY
// On US30 with settings unchanged G1 scores PF 1.472 against E's 1.469 -
// NEUTRAL, not confirmed. Every previously adopted change showed a clear
// same-sign gain on the second instrument; this one does not. It also
// makes 2021 marginally worse (PF 0.89 against E's 0.94).
// About 85 configurations were searched to find a 3% profit-factor gain,
// which is close to what noise alone produces at that search count.
// TRADE E IF YOU WANT THE SETTLED VERSION; TRADE G1 IF YOU ACCEPT A
// THINNER EVIDENCE BASE FOR A BETTER MEASURED CURVE.
// =====
//@version=6
strategy("Gold Trend G1 - regime-scaled risk",
  overlay = true,
  initial_capital = 10000,
  currency = currency.USD,
  default_qty_type = strategy.cash,
  default_qty_value = 1000,
  commission_type = strategy.commission.cash_per_contract,
  commission_value = 0.07,
  // SLIPPAGE IS IN TICKS, NOT POINTS. XAUUSD/OANDA quotes three decimals,
  // so mintick is 0.001 and 200 ticks = 0.20 points - a realistic retail
  // gold fill. The earlier value of 5 modelled 0.005 points, roughly 40x
  // too little, and it flattered every backtest run with it.
  // COST STRESS (full period, Balanced, measured 2026-08-02):
  // 0.05 pt ( 50) -> PF 1.751 +2,533% drawdown 27.7%
  // 0.20 pt (200) -> PF 1.641 +1,620% drawdown 33.4% <- shipped
  // 0.50 pt (500) -> PF 1.452 +715% drawdown 52.2%
  // 1.00 pt (1000)-> PF 1.141 +109% drawdown 76.1%
  // 1.50 pt (1500)-> PF 0.888 -54% BREAK-EVEN IS HERE
  // Set this to YOUR broker's actual gold spread before trusting any
  // number this script reports.
  slippage = 200,
  process_orders_on_close = true,
  // BUG-012 guard. Pine's margin default of 100% silently REJECTS
  // risk-sized orders with no error and no log line. It has bitten
  // this project twice.
  margin_long = 5,
  margin_short = 5,
  calc_on_every_tick = false,
  // REQUIRED by the add-to-winner block. With Pine's default of 0,
  // strategy.entry while a position is open is silently ignored.
  pyramiding = 5,
  // The regime EMA plus its slope lookback reaches ~2,200 bars back.
  // Pine's default buffer is 300, which yields na without an error.
  max_bars_back = 5000)

// — @ Auto-scale —
// Every lookback here is in BARS, so changing timeframe silently doubles
// or halves it. That caused two failed reproductions, so the script
// derives its own settings from the chart instead.
// 15m -> entry 48, SMA 768, EMA 2016, slope 192, trail 6.00
// 30m -> entry 24, SMA 384, EMA 1008, slope 96, trail 4.24
bool autoScale = input.bool(true, "Auto-scale settings to the chart timeframe", group="@ Auto-scale",
  tooltip="ON is strongly recommended. The inputs below are then IGNORED and derived from the timeframe, so moving
  between 15m and 30m cannot silently change the strategy. The dashboard shows what is actually in use. RUN THIS
  ON 30m: it beat 15m on profit factor, return AND drawdown in both halves.")

// — ● RISK PROFILE (the only dial most people need) —
// Same engine, same entry, same exit. Only how HARD it presses.
// Full 6.7 years, XAUUSD 30m, gap-aware fills:
//
// profile adds risk net maxDD worst yr PF
// Conservative 3 ATR x2 1.0% +660% 17.7% -1.5% 1.626
// Balanced 1.5 ATR x4 1.0% +2,307% 29.8% -6.3% 1.728
// Aggressive 1.5 ATR x4 1.5% +8,627% 42.8% -12.1% 1.769
// Maximum 1.5 ATR x4 2.0% +24,712% 54.5% -22.5% 1.764
//
// All four are positive in all five walk-forward slices and all four lose
// only in 2021. Return roughly triples at each step and so does the pain:
// Maximum spent 2021 down 22.5% with a 45.8% intra-year drawdown. Choose
// by the drawdown you can hold through, not by the return.
string riskProf = input.string("25% DD budget", "Risk setting", options=["25% DD budget", "20% DD budget", "Custom"],
  group="● Risk Profile", tooltip="Risk per trade is SOLVED so the historical maximum drawdown lands on the
  stated budget. It is not a preference dial - the drawdown target is the input and the risk percentage is the
  output.\n\n25% DD budget: risk 0.705%. PF 1.945, net +1,875%, max DD 24.98%, MAR 2.26. Monte Carlo median DD
  25.3%, 21.5% chance of exceeding 30%.\n\n20% DD budget: risk 0.553%. PF 1.906, net +998%, max DD 20.00%, MAR
  2.17. Monte Carlo median DD 20.4%, only 5.0% chance of exceeding 30% - the most trustworthy drawdown figure
```

```

    this project has produced.\n\nA backtest drawdown is NOT an expectation. Reordering the same trades gives a
    median about 1 point worse than the historical path. Size for the Monte Carlo median, not the backtest.")

// — ● Audit findings (2026-07-31 expert pass) —————
// 35 entry x exit combinations were measured after this build shipped.
// NOTHING beat it on the criteria that predict live behaviour (zero
// losing walk-forward slices, one losing year, best return-per-drawdown
// among the zero-loss options). Two real findings came out of it:
int   coolDown = input.int(30, "Cooldown after an exit (bars)", minval=0, maxval=100, group="● Audit Findings",
    tooltip="UPDATED 2026-08-03 from 3 to 24 bars (12 hours on 30m). Re-swept 6-48 under entry-ATR add sizing:
    return/drawdown runs 64/60/72/68/74/78/84/79/58/56/22 for cooldown 6/9/12/16/20/24/28/32/36/40/48, a broad
    plateau from 12 to 32 with the baseline at 62. 24 is the middle of that plateau, not its peak (28 scores
    highest). Confirmed on US30: ret/DD 6.73 vs 4.33.\n\nORIGINAL NOTE: after an exit the shipped build could re-
    enter on the very next bar. Swept 0/3/10/24/48 bars: 3 was the free improvement - full period +2,432% to
    +2,533% and drawdown 28.55% to 27.72% with no cost anywhere else. 24 bars gives the best out-of-sample profit
    factor (2.098) but less return; 48 costs a third of the return.")
bool  longSlope = input.bool(false, "Longs ALSO require a rising EMA", group="● Audit Findings",
    tooltip="AUDIT FINDING, defaulted OFF because it is a genuine trade-off, not a free win. Shorts must have the EMA
    falling but longs were gated on nothing - an asymmetry that was never tested until now.\n\nWith longs also
    requiring a RISING EMA: profit factor 1.746 to 1.778, out-of-sample 2.065 to 2.179, and the losing year
    DISAPPEARS - 0 of 7 years negative instead of 1 of 7.\n\nThe cost: one of the five walk-forward slices turns
    negative (0/5 becomes 1/5) and drawdown rises 28.55% to 33.78%. It trades a bad year for a bad quarter. Turn
    it on if a losing calendar year is what you cannot tolerate; leave it off if drawdown is.")

// — @ Entry —————
string entryMode = input.string("Donchian breakout", "Entry trigger", options=["Donchian breakout", "HMA cross"],
    group="@ Entry",
    tooltip="Both measured through the identical exit, filters and adds.\n\nDONCHIAN: PF 1.626, +660.2%, drawdown
    17.68%, return/drawdown 37.3.\n\nHMA CROSS: PF 1.743, +540.4%, drawdown 20.9%, return/drawdown 25.9.\n\nHMA has
    the higher profit factor in BOTH halves (train 1.462 vs 1.370, out of sample 2.008 vs 1.879) - it is more
    accurate per trade. Donchian captures more and does it at less drawdown. Neither is strictly better.\n\nAlso
    measured and NOT offered: SMA and EMA crosses. The SMA 200 cross, the usual default everywhere, is the worst
    setting tested - PF 0.920 and -17.0% with no filters.")
int   entryLen = input.int(50, "Breakout lookback (bars)", minval=5, group="@ Entry",
    tooltip="Long when close exceeds the highest high of the prior N bars. Validated at 50 and 100. A timing trigger,
    not an edge.")
int   hmaLen = input.int(200, "HMA length (HMA mode only)", minval=10, group="@ Entry",
    tooltip="Swept 50/100/200/400/800; 200 best by the weaker of the two halves. NOT auto-scaled - it was measured at
    200 on 30m specifically.")
bool  longOnly = input.bool(false, "Long only (leave OFF)", group="@ Entry",
    tooltip="THIS IS A REGIME BET, NOT A SETTING. Long-only wins every rising year and loses in falling ones. In the
    Feb-Jul 2026 fall, long-only returned -5.72% live while long+short returned +7.57%.")

// — ● G1 REGIME RISK SCALING —————
// The ONLY difference from config E. Risk on a NEW ENTRY is multiplied by
// f1 x f2, clipped to [0.5, 2.0]:
// f1 = 1.4 when |close - regime EMA| / ATR > 6.0, else 0.8
// f2 = 1.4 when SMA(ATR,20) / SMA(ATR,200) > 1.05, else 0.6
// i.e. press when the market is BOTH far from its mean AND expanding;
// shrink when it is neither. Both inputs are causal - no lookahead.
bool  useRegimeRisk = input.bool(true, "G1: scale risk by trend strength x volatility expansion", group="● G1 Regime
    Risk")
float trendThr = input.float(6.0, "Trend-strength threshold (ATR from the regime EMA)", minval=0.5, step=0.5,
    group="● G1 Regime Risk")
float expThr = input.float(1.05, "Volatility-expansion threshold (fast ATR / slow ATR)", minval=0.5, step=0.01,
    group="● G1 Regime Risk")
float upMult = input.float(1.4, "Multiplier when the condition holds", minval=1.0, maxval=2.0, step=0.05, group="●
    G1 Regime Risk")
float dnMultT = input.float(0.8, "Trend multiplier when it does not", minval=0.3, maxval=1.0, step=0.05, group="●
    G1 Regime Risk")
float dnMultV = input.float(0.6, "Volatility multiplier when it does not", minval=0.3, maxval=1.0, step=0.05,
    group="● G1 Regime Risk")

// — @ Risk and filters —————
float riskPct = input.float(1.0, "Risk per trade (%)", step=0.25, group="@ Risk & Filters",
    tooltip="The direct return lever, and it is not free. Measured: 1% -> +85.4% train at 13.6% drawdown; 2% ->
    +208.8% at 28.6%; 3% -> +367.4% at 41.8%. Return scales roughly linearly and so does drawdown. Choose by the
    drawdown you can sit through.")
float maxLev = input.float(20, "Max leverage (notional cap)", minval=1, group="@ Risk & Filters",
    tooltip="BUG-012 guard. Without a notional cap, risk-based sizing can request more than buying power and
    TradingView skips the order with no error.")
float trailMult = input.float(6.0, "Trailing stop (x ATR, 15m basis)", minval=1.0, step=0.5, group="@ Risk & Filters",
    tooltip="THE key parameter, and the one whose old description was wrong. Wider raises PROFIT FACTOR but LOWERS
    RETURN, because risk-based sizing shrinks the position as the stop widens: trail 20 ATR gives PF 1.659 but
    only +23.5%, against trail 6's 1.243 and +85.4%. 6 on a 15m basis (4.24 on 30m) is return-optimal at tolerable
    drawdown.")
int   atrLen = input.int(14, "ATR length", minval=2, group="@ Risk & Filters")
bool  useSma = input.bool(true, "Fast SMA must agree", group="@ Risk & Filters",
    tooltip="750 bars on 15m, about 8 days. Selected on the training half only.")
int   smaLen = input.int(750, "Fast SMA length (bars)", minval=50, group="@ Risk & Filters")
bool  useSma2 = input.bool(true, "Slow SMA must agree TOO (the confluence layer)", group="@ Risk & Filters",
    tooltip="Requiring BOTH averages lifts full-period PF 1.379 -> 1.421 and cuts drawdown 12.30% -> 9.63%. Return
    barely moved; the drawdown fell by a fifth. It is a risk filter, not a return generator.\n\nIt is the
    AGREEMENT that works, not the slower average: SMA2000 instead of SMA750 is worse (train 1.181 vs 1.289), and
    requiring only ONE of the two is much worse (1.149). The two horizons disagreeing marks the choppy middle of a
    range, and that is what gets skipped.")
int   sma2Len = input.int(2000, "Slow SMA length (bars)", minval=100, group="@ Risk & Filters")
bool  useRegime = input.bool(true, "EMA regime gate", group="@ Risk & Filters",
    tooltip="Only long above the EMA, only short below it.")
int   regimeLen = input.int(2000, "Regime EMA length (bars)", minval=20, group="@ Risk & Filters")
bool  useSlope = input.bool(true, "Shorts only when the EMA is FALLING", group="@ Risk & Filters",
    tooltip="This is what makes the system profitable in a falling market without giving up the trend returns. Price
    sits below a FLAT EMA constantly during uptrend pullbacks; the EMA actually declining is far rarer and is what
    a bear regime looks like. Gating shorts on slope keeps them nearly silent in bull markets, where counter-
    trend shorts are what bleed away the edge.")
int   slopeLen = input.int(200, "Slope lookback (bars)", minval=20, group="@ Risk & Filters")
float shortRisk = input.float(0.75, "Short risk multiplier", minval=0.1, maxval=3.0, step=0.05, group="@ Risk &
    Filters",
    tooltip="Counter-trend trades sized smaller. 0.75 gave the best full-period result; raise toward 1.5 for more
    downside protection at the cost of total return.")

// — @ Add to winners —————
bool  usePyr = input.bool(true, "Add to a winning position", group="@ Add to Winners",
    tooltip="THE BIGGEST RETURN LEVER MEASURED HERE, and it multiplies BOTH sides. Walk-forward over five consecutive
    slices, net/drawdown, without adds vs with 2 adds every 3 ATR:\n\n+38.4/9.2 -> +92.5/16.3\n+19.6/6.7

```

```

-> +23.9/13.5\n +8.8/10.4 -> +11.9/16.6\n +15.1/10.2 -> +33.3/16.8\n +28.4/7.8 -> +88.1/13.5\n\nEvery
slice improves, which is why it ships on. But through RANDOM entries it lifts the median from 6.8% to 25.3%
while taking the spread from 6.8 to 29.5 and the worst seed from -3.5% to -30.2%. It is not an edge. It is
betting more on trades that are already working.")
float pyrAtr = input.float(3.0, "Add every (x ATR of favourable move)", minval=0.5, step=0.5, group="Ⓢ Add to
Winners",
tooltip="3.0 with 2 adds is the moderate setting. 2.0 with 4 adds returns far more (+163% and +168% on the first
and last slices) at drawdowns up to 26.4%.")
int pyrMax = input.int(2, "Maximum adds", minval=1, maxval=4, group="Ⓢ Add to Winners")
float pyrRisk = input.float(1.0, "Each add sized at (x original risk)", minval=0.1, maxval=1.0, step=0.1, group="Ⓢ
Add to Winners")

// — Ⓢ Trailing algorithm —————
string trailMode= input.string("Chandelier (ATR)", "Trailing logic", options=["Chandelier (ATR)", "Donchian low"],
group="Ⓢ Trailing Algorithm",
tooltip="Six algorithms were built and measured: chandelier, Donchian low, EMA, a multiple that tightens with
profit, a give-back-a-fraction trail, and none. Two are worth offering.\n\nCHANDELIER (default): PF 1.635,
+679.9%, drawdown 17.01%, positive in ALL FIVE walk-forward slices.\nDONCHIAN LOW: PF 1.624, +935.9%, drawdown
23.45%, NEGATIVE in one of five (-6.0%).\n\nDonchian returns 38% more, but return per drawdown is a dead heat
(39.9 vs 40.0) - the same efficiency at larger size, with one losing period to sit through.\n\nRejected: EMA
trail (best PF 1.383 at 33.7% drawdown), give-back trail (1.061), stepping trail (1.597, behind both).")
float tightAfter= input.float(15.0, "Tighten the trail after this much favourable move (x ATR at entry)", minval=0,
step=1.0, group="Ⓢ Trailing Algorithm",
tooltip="0 disables. PRIORITY 4 FINDING (2026-08-03). Once a trade has run this far in its favour, the leash
tightens to the value below. This is NOT a tight stop and NOT a profit target: it only ever engages on trades
already deep in profit, and the trade still exits on a trail.\n\nAt 20 ATR -> 2.0: profit factor 1.714 ->
1.834, net +3,855% -> +5,922%, drawdown 32.65% -> 33.80%. Against a leverage-matched baseline at the same
drawdown it returns +5,922% vs about +4,500% with a higher profit factor, so it is not the risk dial in
disguise. Improves the profit factor in 5 of 8 calendar years. Confirmed on US30: PF 1.469 vs 1.351 at
identical drawdown.\n\nSwept 10-30 ATR x 1.5-3.0: 21 of 28 combinations beat the baseline and everything from
18 ATR up is a plateau. 20 is inside it, not its peak.")
float tightTo = input.float(2.0, "...tighten it to (x ATR)", minval=0.5, step=0.25, group="Ⓢ Trailing Algorithm")
int donchN = input.int(45, "Donchian trail lookback (bars)", minval=5, group="Ⓢ Trailing Algorithm",
tooltip="The stop rides the lowest low of the last N bars for longs. Do NOT read the training peak at 40 as
meaningful: out of sample the whole 30-50 range performs and the ordering reverses, which is what a fitted
spike looks like once you test it. 45 is the middle of the out-of-sample plateau, not the training maximum.")

// — Ⓢ TP1 / TP2 (staged profit taking, OFF by default) —————
bool tp1On = input.bool(false, "TP1 - bank part of the position", group="Ⓢ TP1 / TP2",
tooltip="MEASURED 20 WAYS AND IT COSTS RETURN EVERY TIME. Best variant: +234.1% against +253.1% with no target.
Adding TP2 as well: +216.7%. Out of sample the gap widens - PF 1.568 with TP1 against 1.930 without. Even
restricting targets to the counter-trend side fails (+242.6%), the only place in this project where that
asymmetry rule has not held.\n\nWhat it DOES buy is a smoother curve: full-period drawdown 17.01% -> 16.04%,
out-of-sample drawdown 12.79% -> 8.38%. A real trade, just not a free one.")
float tp1Atr = input.float(4.0, "TP1 distance (x ATR)", minval=0.5, step=0.5, group="Ⓢ TP1 / TP2")
float tp1Pct = input.float(25, "TP1 - % of position closed", minval=5, maxval=90, step=5, group="Ⓢ TP1 / TP2")
bool tp2On = input.bool(false, "TP2 - bank another part", group="Ⓢ TP1 / TP2")
float tp2Atr = input.float(12.0, "TP2 distance (x ATR)", minval=1.0, step=0.5, group="Ⓢ TP1 / TP2")
float tp2Pct = input.float(25, "TP2 - % of position closed", minval=5, maxval=90, step=5, group="Ⓢ TP1 / TP2")
bool tpShorts = input.bool(false, "Apply TP1/TP2 to shorts only", group="Ⓢ TP1 / TP2")

// — Ⓢ Pure profit-taking mode (different architecture) —————
string tpMode = input.string("Off", "Fixed-target mode", options=["Off", "Percent of price", "Fixed points"],
group="Ⓢ Profit-Taking Mode",
tooltip="TURNS THIS INTO A PROFIT-TAKING STRATEGY instead of a trend-following one. The trail and the adds are
bypassed: entry, static stop, fixed target, done.\n\nMeasured with a 4 ATR stop and a 4% target: train PF
1.402 / +98.5%, out of sample PF 1.420 / +23.2%, full period PF 1.475 / +185.3% at 23.2% drawdown. Train and
out of sample agree to 0.018 of profit factor - the most stable result in this project.\n\nIt beats the
trailing build on profit factor (1.475 vs 1.444 without adds) and loses badly on drawdown and return. Use it
if you want frequent closed wins; leave it Off if you want the return.\n\nUSE PERCENT, NOT POINTS: gold ran
1450 -> 4100 over the sample, so a fixed 40-point target was 2.7% of price at the start and 1.0% at the end.
Percent scored PF 1.475 against points' 1.160.")
float tpPct = input.float(4.0, "Target - percent of price", minval=0.25, step=0.25, group="Ⓢ Profit-Taking Mode",
tooltip="Swept 1.5/2/3/4%. Bigger is monotonically better on profit factor; 4% is where drawdown stops
improving.")
float tpPoints = input.float(160, "Target - fixed points", minval=1, step=5, group="Ⓢ Profit-Taking Mode")
float tpStopAtr = input.float(4.0, "Static stop for this mode (x ATR)", minval=0.5, step=0.5, group="Ⓢ Profit-Taking
Mode",
tooltip="Swept 2/3/4/6/8 ATR. 4 was best; 2 was catastrophic (43% drawdown) because the target then needs an
unreachable hit rate.")

// — Ⓢ Display —————
bool showStop = input.bool(true, "Plot the stop", group="Ⓢ Display")
bool showBand = input.bool(true, "Plot the entry trigger", group="Ⓢ Display")
bool showDash = input.bool(true, "Show dashboard", group="Ⓢ Display")

// =====
// EFFECTIVE SETTINGS
// Targets: breakout ~12h, fast SMA ~8 days, EMA ~21 days, slope ~2 days.
// The trail scales with the SQUARE ROOT of the timeframe because ATR
// grows roughly as sqrt(time); that rule was validated by sweeping the
// 30m trail directly (4.24 was best on train net, +113.5%).
// =====
// Profile presets override the manual risk and add settings.
float riskE = riskProf == "20% DD budget" ? 0.475 : riskProf == "Custom" ? riskPct : 0.606
float pyrAtrE = riskProf == "Custom" ? pyrAtr : 1.5
int pyrMaxE = riskProf == "Custom" ? pyrMax : 4

float tfMin = timeframe.in_seconds() / 60.0
int entryLenE = autoScale ? math.max(5, int(12.0 * 60.0 / tfMin)) : entryLen
int smaLenE = autoScale ? math.max(50, int(8.0 * 24.0 * 60.0 / tfMin)) : smaLen
int regimeLenE = autoScale ? math.max(20, int(21.0 * 24.0 * 60.0 / tfMin)) : regimeLen
int sma2LenE = autoScale ? math.max(100, int(13.0 * 24.0 * 60.0 / tfMin)) : sma2Len
int slopeLenE = autoScale ? math.max(20, int(2.0 * 24.0 * 60.0 / tfMin)) : slopeLen
float trailMultE = autoScale ? 6.0 * math.sqrt(15.0 / tfMin) : trailMult

// =====
// SIGNALS - every ta.* call unconditional at top level (BUG-004)
// =====
float atr = ta.atr(atrLen)
float donHi = ta.highest(high, entryLenE)[1]
float donLo = ta.lowest(low, entryLenE)[1]
float donLoTrail = ta.lowest(low, donchN)[1]
float donHiTrail = ta.highest(high, donchN)[1]
float regimeMa = ta.ema(close, regimeLenE)

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float trendSma    = ta.sma(close, smaLenE)
float slowSma     = ta.sma(close, sma2LenE)
float hmaVal      = ta.hma(close, hmaLen)
bool  hmaUp       = ta.crossover(close, hmaVal)
bool  hmaDn       = ta.crossunder(close, hmaVal)

// G1 regime multiplier. Uses only closed-bar information.
float trendStr    = atr > 0 ? math.abs(close - regimeMa) / atr : 0.0
float atrFast     = ta.sma(atr, 20)
float atrSlow     = ta.sma(atr, 200)
float volExp      = atrSlow > 0 ? atrFast / atrSlow : 1.0
float gif1        = trendStr > trendThr ? upMult : dnMultT
float gif2        = volExp > expThr ? upMult : dnMultV
float regimeMult  = useRegimeRisk ? math.min(math.max(gif1 * gif2, 0.5), 2.0) : 1.0
bool  emaFalling  = regimeMa < regimeMa[slopeLenE]
bool  emaRising   = regimeMa > regimeMa[slopeLenE]
bool  regimeOkL   = (not useRegime or close > regimeMa) and (not longSlope or emaRising)
bool  regimeOkS   = (not useRegime or close < regimeMa) and (not useSlope or emaFalling)
bool  smaOkL      = not useSma or (not na(trendSma) and close > trendSma)
bool  smaOkS      = not useSma or (not na(trendSma) and close < trendSma)
bool  sma2OkL     = not useSma2 or (not na(slowSma) and close > slowSma)
bool  sma2OkS     = not useSma2 or (not na(slowSma) and close < slowSma)

bool  trigUp      = entryMode == "HMA cross" ? hmaUp : (not na(donHi) and close > donHi)
bool  trigDn      = entryMode == "HMA cross" ? hmaDn : (not na(donLo) and close < donLo)

bool  longSig     = trigUp and regimeOkL and smaOkL and sma2OkL
bool  shortSig    = not longOnly and trigDn and regimeOkS and smaOkS and sma2OkS

// =====
// STATE
// =====
bool  tpOn = tpMode != "Off"

var int barsSinceExit = 0
barsSinceExit := strategy.position_size == 0 ? barsSinceExit + 1 : 0
bool  coolOk = barsSinceExit > coolDown

var float trailStop = na
var float tpStop    = na
var float tpTarget  = na
var float tp1Px     = na
var float tp2Px     = na
var bool  tp1Done   = false
var bool  tp2Done   = false
var float lastAdd   = na
var int   nAdds     = 0
// ATR AT ENTRY, for add sizing. PRIORITY 2 FINDING (2026-08-03): sizing each
// add off the CURRENT ATR shrinks it exactly when volatility is expanding,
// which on this system is precisely when the trade is working. Sizing off the
// entry ATR - equivalently, every add equals the ORIGINAL position size -
// scored PF 1.647 vs 1.623, net +1,996% vs +1,617% and drawdown 32.2% vs
// 36.5%. Against a leverage-matched baseline at the same drawdown it returned
// +1,996% vs +1,094% with a higher PF, so it is not the risk dial in disguise.
var float atrEntry = na
// G1* (adopted 2026-08-05): the regime multiplier that sized the OPENING
// unit, held for the life of the trade so the adds inherit it. Re-reading
// the live multiplier at each add would be intra-trade rebalancing -- a
// different system, and one that was not tested.
var float rmEntry  = na
var float bestPx   = na // best excursion THROUGH THE PREVIOUS BAR (see below)

// =====
// TRAILING STOP
// Ratchets in the trade's favour only, anchored to the PREVIOUS bar's
// extreme so the current bar can never move its own stop and then hit it.
// =====
// Excursion is measured off bestPx, which holds the extreme through the
// PREVIOUS bar only. Using this bar's own high would let the bar tighten its
// own stop and then hit it -- the intrabar-lookahead defect of BUG-019.
float excAtr = strategy.position_size == 0 or na(atrEntry) or na(bestPx) or atrEntry <= 0 ? 0.0 :
    (strategy.position_size > 0 ? bestPx - strategy.position_avg_price
     : strategy.position_avg_price - bestPx) / atrEntry
float effTrail = tightAfter > 0 and excAtr >= tightAfter ? tightTo : trailMultE

if strategy.position_size > 0
    float cand = trailMode == "Donchian low" ? donLoTrail : high[1] - atr * effTrail
    trailStop := na(trailStop) ? cand : math.max(trailStop, cand)
else if strategy.position_size < 0
    float cand = trailMode == "Donchian low" ? donHiTrail : low[1] + atr * effTrail
    trailStop := na(trailStop) ? cand : math.min(trailStop, cand)
else
    trailStop := na
    lastAdd   := na
    atrEntry  := na
    rmEntry   := na
    bestPx    := na
    nAdds     := 0

// =====
// ADD TO A WINNING POSITION
// PORT DELTA: with process_orders_on_close the add fills at THIS bar's
// close, so the enlarged position only meets the trailing stop from the
// next bar. The research engine adds at the trigger price and exposes the
// enlarged size on the same bar, so Python is the stricter of the two.
// =====
if usePyr and not tpOn and strategy.position_size != 0 and nAdds < pyrMaxE and not na(lastAdd)
    float step = atr * pyrAtrE
    bool  isLong = strategy.position_size > 0
    float nxt    = isLong ? lastAdd + step : lastAdd - step
    if isLong ? high >= nxt : low <= nxt
        // G1*: the add carries the ENTRY-TIME regime multiplier. Without
        // this the adds size at base risk and the position drifts away
        // from the risk the regime justified when it was opened.
        float aRisk = (isLong ? riskE : riskE * shortRisk) * pyrRisk * nz(rmEntry, 1.0)

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float aAtr = na(atrEntry) ? atr : atrEntry // entry-ATR sizing
float aQty = math.floor(math.min(strategy.equity * aRisk / 100.0 / (aAtr * trailMultE),
    strategy.equity * maxLev / close - math.abs(strategy.position_size)))
if aQty >= 1
    strategy.entry(isLong ? "L" : "S", isLong ? strategy.long : strategy.short, qty=aQty)
    nAdds := nAdds + 1
    lastAdd := nxt

// =====
// ENTRIES
// =====
bool wantShort = shortSig and not longSig
// NOTE: the multiplier is applied to the OPENING position only. Pyramid adds
// stay at base risk. That asymmetry is what was tested and measured; scaling
// the adds too was not, so it is not shipped.
float effRisk = (wantShort ? riskE * shortRisk : riskE) * regimeMult
float stopDist = atr * trailMultE
float qtyRaw = stopDist > 0 ? strategy.equity * effRisk / 100.0 / stopDist : 0
float qtyCap = strategy.equity * maxLev / close
float qty = math.floor(math.min(qtyRaw, qtyCap))

if longSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("L", strategy.long, qty=qty)
    trailStop := close - stopDist
    tpStop := close - atr * tpStopAtr
    tpTarget := tpMode == "Fixed points" ? close + tpPoints : close * (1.0 + tpPct / 100.0)
    tp1Px := close + atr * tp1Atr
    tp2Px := close + atr * tp2Atr
    tp1Done := false
    tp2Done := false
    lastAdd := close
    atrEntry := atr
    rmEntry := regimeMult
    bestPx := close
    nAdds := 0

if shortSig and coolOk and strategy.position_size == 0 and qty >= 1
    strategy.entry("S", strategy.short, qty=qty)
    trailStop := close + stopDist
    tpStop := close + atr * tpStopAtr
    tpTarget := tpMode == "Fixed points" ? close - tpPoints : close * (1.0 - tpPct / 100.0)
    tp1Px := close - atr * tp1Atr
    tp2Px := close - atr * tp2Atr
    tp1Done := false
    tp2Done := false
    lastAdd := close
    atrEntry := atr
    rmEntry := regimeMult
    bestPx := close
    nAdds := 0

// =====
// STAGED PROFIT TAKING (off by default)
// PORT DELTA: these fill at the bar CLOSE via strategy.close, not at the
// target price. Two live limit orders plus a trailing stop on one entry
// is where order conflicts come from. The research engine fills at the
// target, so expect Pine to read slightly worse here.
// =====
bool tpSideOk = not tpShorts or strategy.position_size < 0
bool tp1Hit = tp1On and not tpOn and not tp1Done and tpSideOk and strategy.position_size != 0 and
    (strategy.position_size > 0 ? high >= tp1Px : low <= tp1Px)
bool tp2Hit = tp2On and not tpOn and not tp2Done and tpSideOk and strategy.position_size != 0 and
    (strategy.position_size > 0 ? high >= tp2Px : low <= tp2Px)

if tp1Hit
    strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp1Pct, comment="TP1")
    tp1Done := true
if tp2Hit
    strategy.close(strategy.position_size > 0 ? "L" : "S", qty_percent=tp2Pct, comment="TP2")
    tp2Done := true

// =====
// EXITS
// Fixed-target mode replaces the ratchet entirely: static stop, fixed
// limit, no trail and no adds. Anything else would be a third
// architecture that was never measured.
// =====
if tpOn
    if strategy.position_size > 0 and not na(tpStop)
        strategy.exit("L-exit", "L", stop=tpStop, limit=tpTarget)
    if strategy.position_size < 0 and not na(tpStop)
        strategy.exit("S-exit", "S", stop=tpStop, limit=tpTarget)
else
    if strategy.position_size > 0 and not na(trailStop)
        strategy.exit("L-exit", "L", stop=trailStop)
    if strategy.position_size < 0 and not na(trailStop)
        strategy.exit("S-exit", "S", stop=trailStop)

// Roll bestPx forward only now that this bar's stop and exits are resolved,
// so the value the next bar reads is the extreme through THIS bar and no
// bar can ever act on its own high.
if strategy.position_size != 0
    bestPx := na(bestPx) ? close : (strategy.position_size > 0 ? math.max(bestPx, high) : math.min(bestPx, low))

// =====
// VISUALS
// =====
plot(showBand and entryMode == "Donchian breakout" and not na(donHi) ? donHi : na, "Breakout high",
    color.new(color.orange, 40), 1, plot.style_stepline)
plot(showBand and entryMode == "Donchian breakout" and not na(donLo) ? donLo : na, "Breakout low",
    color.new(color.orange, 70), 1, plot.style_stepline)
plot(showBand and entryMode == "HMA cross" ? hmaVal : na, "HMA", color.new(color.fuchsia, 0), 2)
plot(showStop and strategy.position_size != 0 ? (tpOn ? tpStop : trailStop) : na, "Stop", color.new(#D50000, 0), 2,
    plot.style_stepline)
plot(tpOn and strategy.position_size != 0 ? tpTarget : na, "Fixed target", color.new(#00E676, 0), 2,

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        plot.style_stepline)
plot(tp1On and not tpOn and strategy.position_size != 0 and not tp1Done ? tp1Px : na, "TP1", color.new(#00E676, 20),
    1, plot.style_circles)
plot(tp2On and not tpOn and strategy.position_size != 0 and not tp2Done ? tp2Px : na, "TP2", color.new(#00E676, 45),
    1, plot.style_circles)
plot(useRegime ? regimeMa : na, "Regime EMA", color.new(color.fuchsia, 55), 1)
plot(useSma ? trendSma : na, "Fast SMA", color.new(color.aqua, 25), 2)
plot(useSma2 ? slowSma : na, "Slow SMA", color.new(#7E57C2, 15), 2)

plotshape(longSig and strategy.position_size == 0, "Long entry", shape.triangleup, location.belowbar,
    color.new(#00C853, 0), size=size.small, text="LONG")
plotshape(shortSig and strategy.position_size == 0, "Short entry", shape.triangledown, location.abovebar,
    color.new(#D50000, 0), size=size.small, text="SHORT")

bool exitedLong = strategy.position_size[1] > 0 and strategy.position_size == 0
bool exitedShort = strategy.position_size[1] < 0 and strategy.position_size == 0
plotshape(exitedLong or exitedShort, "Exit", shape.xcross, location.absolute, color.new(color.gray, 0),
    size=size.tiny)

// =====
// ALERTS
// =====
alertcondition(longSig, "Gold Trend - LONG entry", "LONG {{ticker}} {{interval}} at {{close}}")
alertcondition(shortSig, "Gold Trend - SHORT entry", "SHORT {{ticker}} {{interval}} at {{close}}")
alertcondition(exitedLong or exitedShort, "Gold Trend - position closed", "EXIT {{ticker}} {{interval}} at {{close}}")
alertcondition(tp1Hit, "Gold Trend - TP1 hit", "TP1 {{ticker}} {{interval}} at {{close}}")
alertcondition(tp2Hit, "Gold Trend - TP2 hit", "TP2 {{ticker}} {{interval}} at {{close}}")

// =====
// DASHBOARD
// =====
var table dash = table.new(position.top_right, 2, 10, bgcolor=color.new(color.black, 20), border_width=1,
    border_color=color.new(color.gray, 60))
if showDash and barstate.islast
    int nT = strategy.closedtrades
    int wT = 0
    float gw = 0.0
    float gl = 0.0
    if nT > 0
        for i = 0 to nT - 1
            float p = strategy.closedtrades.profit(i)
            wT += p > 0 ? 1 : 0
            gw += p > 0 ? p : 0.0
            gl += p < 0 ? -p : 0.0
    float wr = nT > 0 ? wT * 100.0 / nT : na
    float pf = gl > 0 ? gw / gl : na
    float netPct = (strategy.equity - strategy.initial_capital) / strategy.initial_capital * 100
    int nUp = (regimeOkL ? 1 : 0) + (smaOkL ? 1 : 0) + (sma2OkL ? 1 : 0)
    int nDn = (regimeOkS ? 1 : 0) + (smaOkS ? 1 : 0) + (sma2OkS ? 1 : 0)

    table.cell(dash, 0, 0, "GOLD TREND", text_color=color.orange, text_size=size.small)
    table.cell(dash, 1, 0, riskProf + " " + str.tostring(riskE, "#.#") + "%", text_color=color.orange,
        text_size=size.small)
    table.cell(dash, 0, 1, "Trades", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 1, str.tostring(nT), text_color=nT < 250 ? color.yellow : color.white, text_size=size.tiny)
    table.cell(dash, 0, 2, "Win rate", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 2, na(wr) ? "-" : str.tostring(wr, "#.#") + "%", text_color=color.white, text_size=size.tiny)
    table.cell(dash, 0, 3, "Profit factor", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 3, na(pf) ? "-" : str.tostring(pf, "#.##"), text_color=na(pf) ? color.gray : pf >= 1.3 ?
        color.lime : pf >= 1.0 ? color.yellow : color.red, text_size=size.tiny)
    table.cell(dash, 0, 4, "Net", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 4, str.tostring(netPct, "#.#") + "%", text_color=netPct > 0 ? color.lime : color.red,
        text_size=size.tiny)
    table.cell(dash, 0, 5, "Position", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 5, strategy.position_size > 0 ? "LONG +" + str.tostring(nAdds) : strategy.position_size < 0 ?
        "SHORT +" + str.tostring(nAdds) : "flat", text_color=strategy.position_size != 0 ? color.lime : color.gray,
        text_size=size.tiny)
    table.cell(dash, 0, 6, "Stop", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 6, na(trailStop) and na(tpStop) ? "-" : str.tostring(tpOn ? tpStop : trailStop,
        format.mintick), text_color=color.red, text_size=size.tiny)
    table.cell(dash, 0, 7, "Confluence", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 7, nUp == 3 ? "LONG 3/3" : nDn == 3 ? "SHORT 3/3" : "none (" + str.tostring(math.max(nUp,
        nDn)) + "/3)", text_color=nUp == 3 ? color.lime : nDn == 3 ? color.red : color.gray, text_size=size.tiny)
    table.cell(dash, 0, 8, "Exit engine", text_color=color.silver, text_size=size.tiny)
    table.cell(dash, 1, 8, tpOn ? "fixed target" : trailMode == "Donchian low" ? "donchian " + str.tostring(donchN) :
        "chandelier " + str.tostring(trailMultE, "#.##"), text_color=color.aqua, text_size=size.tiny)
    table.cell(dash, 0, 9, autoScale ? "auto " + timeframe.period : "MANUAL", text_color=autoScale ? color.aqua :
        color.orange, text_size=size.tiny)
    table.cell(dash, 1, 9, autoScale ? "sma " + str.tostring(smaLenE) + "/" + str.tostring(sma2LenE) : (nT < 250 ?
        "n<250: unstable" : ""), text_color=color.silver, text_size=size.tiny)

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